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Mapping Individual Voice Quality over the Voice Range - the Measurement Paradigm of the Voice Range Profile

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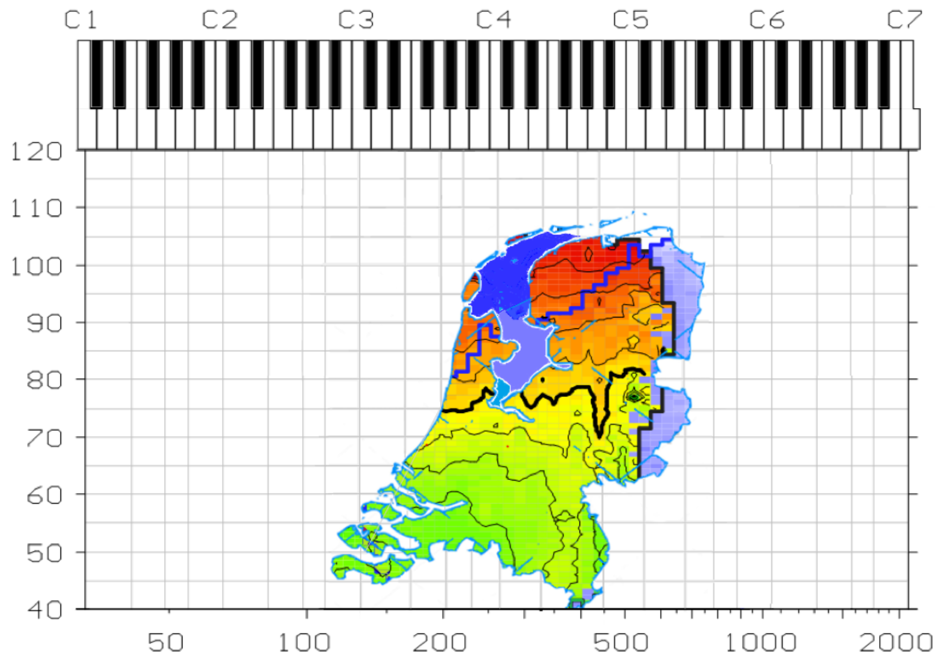

The cover graphic: Voice map of the Netherlands, created by Sten Ternström as an amusement for his G. Paul Moore lecture 'Mind the Gap' at the Annual Symposium of the Voice Foundation in 2017. The graphics were further improved upon by Marjolein Pabon.

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Abstract

The acoustic signal of voiced sounds has two primary attributes: fundamental frequency and sound level. It has also very many secondary attributes, or ‘voice qualities’, that can be derived from the acoustic signal, in particular from its periodicity and its spectrum. Acoustic voice analysis as a discipline is largely concerned with identifying and quantifying those qualities or parameters that are relevant for assessing the health or training status of a voice or that characterize the individual quality. The *thesis* presented here is that all such voice qualities co-vary essentially and individually with the fundamental frequency and the sound level, and that methods for assessing the complete voice must account for this co-variation and individuality. The five studies point to ways of doing this in practice. The central theme is a voice quality mapping paradigm that relies on an interactive charting procedure. On such a map is represented one of several acoustic parameters that describes a particular quality aspect of the voice. In its bare form, the map sketches only the contour of the sound intensity range as a function of the tonal range; a representation known as a 'phonetogram', or 'voice range profile'. In its extended form, the map shows voice parameter distributions as color gradients and contour lines, and the mapping area is called the 'voice field'. The method builds on the empirical finding that the fundamental frequency (perceived as pitch) and the sound pressure level (connected to vocal effort) are prescriptive for the quality of an individual voice. Revisiting a location on the map will yield a reproducible quality measurement, and it becomes possible to navigate over the map to a specific individual quality setting. The recurrent topics are: how might such maps be interpreted, in order to reveal the underlying mechanism? How might an optimal and comprehensive parametric representation be derived? How does the proportional scale mapping, emblematic for the mapping paradigm, relate to a physiological interpretation of the results? Is there an overarching principle?

–Article A is a clinically oriented 'proof of concept' pilot study, which introduces the recording method and the principle of mapping several voice quality parameters in parallel. Early on, this study shows that the hope of assessing objectively the quality or health status of a voice lies in distribution shapes, rather than in any particular measurement.

–In Article B, the recording technology has improved: the voice parameters are less noisy and more specific, and more recordings have been made. Although the distribution patterns are reproducible, it remains difficult to interpret the individual maps. Some first attempts are made to sketch a general picture of how voice quality changes over the voice field.

–From Article C, the phonetogram continues under the name voice range profile (VRP). An interpolation model based on the Fourier transform is introduced. With this model, the curves that outline the individual contours are resampled to match a common base, and can next be averaged. A decisive feature of this interpolation model is that the variation that occurs along each of the two axes is now seen as an effect of one and the same factor. The model thus conforms to the idea that the specific shape of the VRP curve and the connected quality pattern are expressions of a common underlying physiologic apparatus. A novel finding was that the VRP curves tend to align in the northwest corner. There, per voice group, the individual voices converged on one coherent limit.
–Article D reports on a large longitudinal study on the effect of voice change as a result of voice training. The voice maps of the singing students, who were followed over the four years of their BA studies, show large individual differences in shape and quality distribution. Only for a group of 10 classically trained female students, isolated from the group of 115 participating students, was it possible to document a training effect. The voice quality change appeared as a change in the distribution pattern of the crest factor. The effect could be attributed to the training of mixing as a vocal technique, and to the avoidance of the use of pushed-up chest voice.
–In Article E the individual boundaries of the vocal range are irrelevant due to the averaging technique used; the focus is solely on spectral changes over the voice field. The goal is to compile an atlas of the voice, with acoustic feature maps per voice register, for three voice groups; trained and untrained female voices, and trained male voices. Feature maps are sketched for eight voice parameters, each representing a different aspect of the spectral envelope. With this atlas in hand, a number of fundamental principles that control the acoustics of voice production are discussed. The second aim of article V is to demonstrate the large differences between voices, in the way the spectrum varies over the voice field.
–Article F is provided for convenience as an introductory tutorial to the recording method, and to the standard applications of the VRP. The inclusion of voice quality parameters is mentioned, but not elaborated upon. This overview article is not part of the scientific contribution of the present thesis.
–The framework text presents the theoretical basis for the analysis model in relation to the practical principles. The central interest in the VF recording paradigm is to map the *proportional* dependencies that exist between voice parameters. These proportional dependencies are characterized by shared exponential factors, which are exposed by mapping under a pervasive logarithmic scaling. The ability to record over a maximum exponential scale range determines the design of the recording system. The recording paradigm conforms to a maximum entropy criterion, which has its own specific statistics. This criterion

presents an optimization principle for individual voice parameters, and maximizes the probability of revealing a possible connection between voice parameters, given the total information content observed. The recurring outcome is that voice metric distribution shapes differ considerably between voices, but are consistent within a voice. This in-voice consistency makes it possible to navigate on the map of an individual voice, while the differences between voices make it difficult to derive a general picture by averaging. Using a thermodynamic model of a 'balloon in a box', a fundamental principle from statistical mechanics is described that, when applied to the voice, explains why voice parameter distribution patterns are only qualitatively comparable. Absolute scales and absolute positions, with which we are so familiar in experimental research, are more a hindrance than a help when studying a phenomenon so variable as human voice production. There is no guarantee that all voices will work from the same point of reference and will scale up in the same manner. It is precisely in this dynamic upscaling that the individuality of the voice is apparent.

Keywords

Voice Range Profile (VRP), Phonetogram, Voice Field, Voice quality, Singing voice, Voice synthesis, Clinical voice assessment, Sound level measurement, Maximum Entropy.

Sammanfattning

Den akustiska signalen för tonande röstljud har två primära attribut: fonationsfrekvens och ljudnivå. Den har också en mångfald sekundära attribut, eller 'röstkvaliteter', som kan härledas ur den akustiska signalen, och då särskilt ur dess periodicitet och dess spektrum. Akustisk röstanalys syftar vanligen till att identifiera och kvantifiera sådana röstkvaliteter eller akustiska mått som är relevanta för bedömning av röstens status, vad gäller t ex hälsa eller träning. Den här framlagda tesen säger att alla sådana röstkvaliteter samvarierar, essentiellt och individuellt, med fonationsfrekvens och ljudnivå, och att metoder för bedömning av rösten i dess helhet måste beakta denna samvariation och individualitet. De fem delstudierna pekar på hur detta kan göras i praktiken.

Det centrala temat är ett paradigm för kartläggning av människoröstens akustiska egenskaper, nära förknippat med en procedur för interaktiv inspelning. En röstkarta visar hur något givet röstmått varierar över ljudtrycksnivå och fonationsfrekvens. I sin grundläggande form visar kartan endast hur gränserna för röstomfånget i ljudnivå (svagast och starkast möjliga röst) varierar med fonationsfrekvensen, en form som kallas *fonetogram* eller *röstomfångsprofil*. I en utökad form används färger och isokonturer på kartan för att visa hur givna röstmått varierar även däremellan; en sådan form kallas *röstfält*. Empiriska iakttagelser visar att fonationsfrekvensen (uppfattad som tonhöjd) och ljudnivån (förknippad med upplevd röststyrka) är avgörande för röstkvaliteten, på ett sätt som är specifikt för individen. Upprepade mätningar på samma individ vid samma läge på kartan ger reproducerbara resultat, och det blir möjligt att med kartans hjälp navigera till givna individspecifika inställningar av röstkvaliteten. Vissa frågor återkommer envist: hur dessa kartor ska tolkas, för att belysa de underliggande röstproduktionsmekanismerna? Hur bör man utforma en optimal och heltäckande representation av röstens funktion? Mätparadigmet föreskriver genomgående en proportionell snarare än linjär skalning, men hur förhåller sig en sådan till en fysiologisk tolkning av resultaten? Finns det någon övergripande princip?

– Delstudie A är ett kliniskt inriktat ”proof of concept”. Här presenteras inspelningsmetoden, liksom principen att visualisera flera olika röstmått parallellt. Redan denna tidiga studie visar att relevansen hos objektiva röstmått måste ligga i deras fördelning över röstomfånget, snarare än i enstaka stickprov.

– I delstudie B har tekniken förbättrats avsevärt, och fler inspelningar har gjorts; röstmåtten är också mindre brusiga och tydligare. Fördelningarna är reproducerbara, men kartorna är fortfarande svåra att tolka. Några inledande försök görs att skissa en mer allmängiltig bild av hur röstmåtten varierar över röstomfånget.

– I delstudie C ersätts termen ’fonetogram’ av ’röstomfångsprofil’. En Fourier-modell föreslås för interpolering och omsampling av konturkurvor till ett enhetligt format. Därmed kan röstomfångsprofiler jämföras mellan studier, och medelkurvor kan beräknas. En viktig egenskap hos modellen är att samvariationer längs axlarna nu tydligare avbildas som en verkan av en gemensam underliggande orsak. Modellen underbygger sålunda tanken att röstomfångskurvans form och därmed sammanhängande röstmåttsfördelningar avspeglar en och samma fysiologiska mekanism. Ett intressant rön är att röstomfångskurvor från olika individer inom samma grupp tenderar att sammanfalla i omfångets ’nordvästra’ del.
– Delstudie D är en stor longitudinell studie av hur röstanvändningen förändrades under en fyraårig sångutbildning. Röstkartorna påvisade stora individuella förändringar i röstomfångets form och i röstmåttens fördelningar under utbildningens gång. 115 sångstudenter deltog, men det var först när urvalet skalades ned till tio kvinnliga studenter av klassisk sång som gruppen blev så pass enhetlig att effekter av utbildningen gick att belägga. Effekterna, som yttrade sig i ändrade fördelningar hos ljudsignalens s.k. toppfaktor, kunde tillskrivas uppträdnade registerövergångar och undvikande av forcerad modalröst (’bröstregister’).
– I delstudie E är de individuella rösternas omfång irrelevant; fokus är istället på att få fram gruppmedelvärden av de spektrala förändringar som uppstår över röstfältet. Syftet är att framställa en ’atlas’ över rösten, med separata kartor för modal- och falsettröst (motsvarande bröst/huvudklang), för tre olika grupper: kvinnor med och utan röstträning, och rösttränade män. Kartor uppvisas över åtta olika röstmått, som var och en beskriver en enskild aspekt av spektrum-enveloppen för vokalen /a/. Med denna kartbok i hand diskuteras ett antal grundläggande principer för ljudproduktionens akustik. Ett ytterligare syfte är att presentera en realistisk bild av hur olika röster kan vara, när det gäller hur spektrum förändras över röstens omfång.

Artikel F är ingen delstudie, utan en metodöversikt. Den bilägges här som en allmän introduktion till inspelningsmetoden och som en redogörelse för vanliga tillämpningar av röstomfångsprofilen. Utökningen till röstmått över röstfältet liksom dessas uttolkning behandlas endast summariskt. Denna artikel inräknas inte i avhandlingens vetenskapliga bidrag.

Ramberättelsen presenterar analysmodellens teoretiska grund och relaterar den till de praktiska principerna. Det väsentliga i analysparadigmet är att kartlägga de *proportionella* (till skillnad från *linjära*) beroenden som bör finnas mellan röstmåtten. Varje konstant proportionellt beroende beskrivs av en konstant exponent, och den senare kan lättast bestämmas genom att man konsekvent använder en logaritmisk skalning i alla dimensioner. Inspelningssystemet utformades därför omsorgsfullt för att tillåta en så bred variation som möjligt i exponenterna. Inspelningsparadigmet följer ett kriterium för maximal entropi, som lyder under en särskild beskrivande statistik. Detta kriterium leder till en optimeringsprincip för individuella röstparametrar, och maximerar sannolikheten att finna samband mellan olika röstparametrar, givet den tillgängliga informationen. Gång på gång visar det sig att röstmåttens fördelningar skiljer avsevärt mellan röster, men är reproducerbara för en given röst. Denna reproducerbarhet inom individer gör det möjligt att orientera sig på individens röstkarta, medan skillnaderna mellan individer gör det svårt att skapa en mer allmängiltig bild genom t ex medelvärdesbildning. Med hjälp av en termodynamisk modell av en ballong i en låda beskrivs en grundprincip från statistisk mekanik. När den tillämpas på rösten, förklarar den varför röstmåttens fördelningar endast är kvalitativt jämförbara. Absoluta skalor och mätetal, som vi är vana vid från experimentell forskning, är mer till hinder än till hjälp när vi studera något så variabelt som människorösten; det finns ingen garanti för att referensvärden och skalfaktorer ska vara desamma från en röst till en annan. Det är just i denna dynamiska skalning som röstens individualitet ger sig till känna.

Samenvatting

Het akoestische signaal van de stem kent twee primaire kenmerken: de grondtoonfrequentie en het geluidsniveau. Daarnaast is er een reeks van secondaire attributen, of 'stemkwaliteitskenmerken', die afgeleid kunnen worden van het akoestische signaal, met name uit de periodiciteit van het signaal en uit het spectrum. Bij akoestische stemkwaliteitsanalyse gaat het om het identificeren en kwantificeren van kenmerken of parameters die iets zeggen over de gezondheid van de stem of de mate van stemtraining, maar ook over de vraag welke kenmerken eigen zijn aan een stem. De *thesis* die hier wordt gepresenteerd is, dat de akoestische stemkenmerken co-variëren met de grondtoonfrequentie en het geluidsniveau op een manier die per stem verschilt, en dus moet voor de bepaling van de individuele stemkwaliteit of stemkwaliteitsverandering, de gehele co-variatie over het stembereik in ogenschouw worden genomen. De vijf studies beschrijven hoe dit in de praktijk gedaan kan worden.

Het centrale thema is een specifieke manier om stemkwaliteit vast te leggen, die gebaseerd is op het interactief tekenen van een landkaart van de stem. Uit een reeks van akoestische kenmerken, die ieder een ander stemkwaliteitsaspect representeren, wordt er telkens één op de kaart afgebeeld. In zijn kale vorm laat de kaart alleen de omtrek van de stem zien: het intensiteitsbereik als functie van het toonbereik, en wordt dan een 'fonetogram' of 'voice range profile' genoemd. In zijn uitgebreide vorm worden de verdelingen van de stemkwaliteitsparameters met een kleurgradiënt en hoogtelijnen afgebeeld op wat nu het 'stemveld' genoemd wordt. De meting is gebaseerd op het empirische principe dat de toonhoogte en het geluidsniveau bepalend zijn voor de kwaliteit van de individuele stem. Door de plaatsbepaling op de kaart wordt het mogelijk naar een specifieke steminstelling terug te keren. De stemmeting wordt hierdoor reproduceerbaar. Terugkerende vragen zijn: hoe deze kaarten te interpreteren, wat is het onderliggende mechanisme? Hoe kunnen we een optimale en zo compleet mogelijke beschrijving krijgen? Op welke wijze is de proportionele (logaritmische) schaal, die kenmerkend is voor dit afbeeldingsconcept, een afspiegeling van een algemeen fysiologisch principe? Is er een overkoepelend mechanisme te ontdekken?

–Artikel A is een klinisch georiënteerde 'proof of concept' pilotstudy en introduceert de methodiek rond de automatische stemopname en het principe van het naast elkaar afbeelden van verschillende stemkwaliteitskenmerken. Al vanaf het begin maakt deze studie duidelijk dat de toekomst van de objectieve stemkwaliteitsmeting ligt bij het bepalen van een stemkwaliteitsverdeling en veel minder bij het doen van slechts een enkele meting.

–Bij Artikel B is de opnametechnologie sterk verbeterd en zijn er meer stemregistraties gemaakt; ook zijn de parameters zuiverder en eenduidiger. De verdelingen zijn reproduceerbaarder, maar de kaarten blijven moeilijk te interpreteren. Door binnen een stem de samenhang te zoeken tussen de verdelingspatronen voor de verschillende stemparameters, wordt geprobeerd het onderliggende mechanisme te herleiden. Ook komt het probleem van het middelen van stemkwaliteitspatronen aan de orde en wordt een eerste poging wordt gedaan om door middeling een beeld te schetsen hoe in het algemeen de stemkwaliteit verandert over het stemveld.
–Vanaf Artikel C gaat het fonetogram verder onder de naam voice range profile (VRP). Er wordt een op de Fouriertransformatie gebaseerd interpolatiemodel geïntroduceerd. Daarmee wordt het mogelijk om de contouren, die de individuele stemomvang schetsen, allemaal met eenzelfde aantal knikpunten te beschrijven en zo puntsgewijs te middelen. Bepalend voor dit interpolatiemodel is, dat de variatie die langs ieder van de twee assen optreedt, als één beweging wordt gezien. Het model sluit zo aan bij het later verder uitgewerkte inzicht, dat de specifieke vorm van de contour en het kwaliteitspatroon een uiting zijn van hetzelfde onderliggende principe. Met dit interpolatiemodel werden de gemiddelde contourcurves die voor verschillende stemgroepen in de literatuur te vinden zijn, objectief vergeleken. Deze standaardcurves, die per groep elkaar, theoretisch gezien, zouden moeten overlappen, lopen echter ver uit elkaar. De grote verschillen worden verklaard door de aanzienlijke verschillen in opnameapparatuur en opnamemethodiek. Een nieuwe ontdekking is dat de noordwesthoek in het stemveld bij herhaling de locatie is waar de stemmen binnen een stemgroep convergeren naar dezelfde grens.
–Artikel D beschrijft het resultaat van een groot longitudinaal onderzoek naar het effect van stemverandering ten gevolge van stemtraining. De stemkaarten van de zangstudenten, die vanaf het begin van hun studie tot aan hun afstuderen vier jaar lang gevolgd werden, laten onderling grote individuele verschillen zien in vorm en stemkwaliteitsverdeling. Van de 115 deelnemende studenten lukt het maar voor een groep van 10 klassiek geschoolde, vrouwelijke studenten om een vergelijkbaar effect te isoleren. De stemkwaliteitsverandering die wordt gemeten laat zich zien als een wijziging in de verdeling van de crest factor. Deze verandering is te verklaren door het nieuw aangeleerde gebruik van registermenging als zangtechniek en in het vermijden van het gebruik van de opgedreven borststem. Opvallend is dat de stemtraining in zeer geringe mate leidt tot een vergroting van het toonbereik, terwijl voor deze groep niet een toename, maar een significante afname in geluidsniveau wordt gezien in het middengebied van het stembereik.

–In Artikel E spelen de individuele begrenzingen van het stembereik door de gebruikte middelingstechniek geen rol en zijn hier geen aandachtspunt. De focus ligt op de spectrale veranderingen die zich over het stemveld voltrekken. Het doel is een atlas van de stem samen te stellen, met stemkaarten per stemregister, voor drie verschillende stemgroepen: getrainde- en ongetrainde vrouwenstemmen en getrainde mannenstemmen. Er zijn deelkaarten voor acht verschillende parameters, die elk een ander facet van de spectrale contour weergeven. Met deze atlas in de hand wordt een aantal fundamentele principes, die de akoestiek van de stemproductie beheersen, bediscussieerd. Het tweede doel van dit artikel is een realistisch beeld te schetsen van grote verschillen tussen stemmen onderling, in de wijze waarop het spectrum verandert over het stemveld. Slechts oppervlakkig gezien lijken de verdelingspatronen op elkaar.
–Artikel F is toegevoegd als algemene introductie op de opnamemethodiek en beschrijft de standaardtoepassing van de VRP en uitbreidingen daarop. Het toevoegen van kwaliteitsparameters en de interpretatie van de patronen komen slechts zijdelings aan de orde. Dit overzichtsartikel maakt geen onderdeel uit van de wetenschappelijke bijdragen voor dit proefschrift.
–De verbindende tekst beschrijft de theoretische basis van het analysemodel en de achterliggende praktische principes. Het centrale doel bij het stemopnameconcept is het in kaart brengen van de proportionele afhankelijkheid die er tussen stemparameters bestaat. Een constante proportionele afhankelijkheid betekent een constante exponentiële factor. Daarom zijn alle schalen logaritmisch. Het kunnen meten over een zo groot mogelijk exponentieel bereik bepaalt het ontwerp van het opnamesysteem. Het opnameconcept volgt een 'maximum entropie criterium' (d.w.z. er is geen ordeningsprincipe waaraan voldaan moet worden); hierbij hoort een eigen specifieke statistiek. Dit criterium levert een optimalisatieprincipe voor individuele stemparameters en garandeert de beste conditie om een mogelijk onderling verband tussen stemparameters uit de beschikbare informatie te halen. De terugkerende observatie blijft dat stemkaarten tussen stemmen onderling sterk verschillen en dat moeilijk het is om met middeling een beeld te schetsen van de wijze waarop stemkwaliteit in het algemeen varieert. Aan de hand van een thermodynamisch model van een 'ballon in een doos', wordt een principe uit de statistische mechanica uitgelegd, dat, wanneer toegepast op de stem, verklaart, wat de achterliggende reden is waarom verdelingspatronen alleen oppervlakkig gezien vergelijkbaar zijn. Een absolute schaal en absolute positionering geven geen garantie dat alle stemmen vanuit hetzelfde referentiepunt werken en op dezelfde wijze opschalen. Juist in de dynamische opschaling komt de individualiteit van de stem naar boven.

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List of publications

Paper A

Pabon JPH, Plomp R (1989). Automatic phonetogram recording supplemented with acoustic voice quality parameters. J Speech and Hearing Res, 31, 710-722.

Paper B

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Paper C

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1 CONTENTS

PREFACE

Individual voice quality

What makes sound so special? What makes a voice sound so special? It has always been my dream (or obsession) to catch and control the acoustic feature which makes a voice unique, to point out that specific signal feature which distinguishes one particular voice from all others. Individual voice quality is definitely a personal fascination.

Still, it is not this personal fascination alone that the term 'individual' appears in the title of this thesis. Measuring the voice is modelling the voice. We understand the voice through our models. We are able to communicate our scientific findings by the abstractions that we make. Each voice exhibits many facets of quality, where each abstraction or parameterization that aims at quantifying such a facet will represent only a snippet of the total reality. Within these fragmented representations reality, there is generally little room for characterizing an individual quality in a comprehensive way. In most scientific studies, a large number of observations N is strived for, and the voice is only one of many. Any abstraction is essentially a reduction, and consensus appears to say that it is impossible to be complete: it is unrealistic given the time, the money, the scope. The abstraction, the clever reduction is the ideal that demonstrates the triumph of science and the intellect of its inventor. In research that subscribes to such an ambition, the individual's voice quality becomes, at the most, a single point in a multidimensional space, where only a blatant deviation from an average might explain why a particular voice stands out.

This thesis is concerned with counterbalancing the preoccupation with reduction. It breathes the idea that it is possible to be comprehensive yet specific on a single voice. The averaged voice has enough support in science; it is the individual voice that needs special attention and support. It just needs its own protected area—the voice field—where it is permitted to fully express its true qualities.

Instrumental component

Any objective knowledge that can be collected on the quality of the voice is essentially linked to a model. In the recording of voice quality, the implementation of the model plays a crucial role. Regardless of how much we approve the model, *a model can never convey more truth than its implementation*, assuming that it is reality that defines our truth. The implementation remains the wrapper around the model.

This standpoint summarizes a general tone of this work. It reflects the aim to stay close to reality, and to push the borders of what is known by pushing the borders of the implementation.

Thus, "the voice recording system", the materialization of the model, will play a prominent role in this text. Progressively, the development of knowledge became synonymous with the successive development of more advanced voice recording systems. A line that reflects in the successive papers. At a certain point, the intricate questions about the implementation of the model and the interpretation of the recorded results came to mirror the search for answers to some very fundamental scientific questions on voice production. This notion brought the extra motivation to seriously enter the scientific arena and to report maturely on these answers and questions. Once the mapping paradigm has been introduced in chapter 1, chapter 2 will retrospectively sketch the historical context. Fragments with personal opinions and angles will pop up in this chapter, where I hope that the reader and I share the scientific interest.

> At several places in the framework text, the margins will narrow, to inform the reader of a change in narrator perspective. Here the scientist steps back, to let the recording system developer, the salesman or the performer have his say. Sometimes the scientific perspective is best formulated from this alternative viewpoint.

Structure of the framework text

This thesis combines results of work done over a period of more than 35 years. There are simply too many results and answers to report on, that again relate to questions and viewpoints that reflect on a context -the voice field- that is not generally explored in this manner. The attached peer-reviewed papers (**A…E**) report only a small portion of the many answers and questions encountered along the way; typically, the more established parts. The wording is intentional: the *answers* preceded the *questioning*. In fact, this framework text is structured by formulating questions that connect to answers that were first brought to light by the voice recording system. The mere act of interpreting the voice maps can be seen as a form of reverse-engineering of the acoustics of the voice production system. In analogy with these inversions, this text lacks a conclusive chapter (if the introductory chapter is ignored). As all questions stem from earlier conclusions and interpretations, and not from external sources, it would be strange to pretend that a series of well formulated research questions formed the actual starting point of this thesis.

Short synopsis

- The introduction (chapter 1), explains the mapping paradigm. It concludes by presenting a complex of questions around the re-enacted experience that is considered emblematic for the voice field mapping paradigm: The specificity and reproducibility that can be reached in mapping individual voice quality, which also seems to imply a repudiation of generality when different voices need to be compared.
- Chapter 2 sketches in an anecdotal way the historical context. The sometimes-detached personal perspective is exposed.
- Chapter 3 introduces the concept of exponential sampling that gives the voice field its logarithmic base. The same exponential sampling defines the design and setup of the recording system and facilitates the perceptual parallels.
- Chapter 4 discusses what changes when the statistical variance is assessed on a logarithmic scale. It explains when and where this entity actually refers to an exponentially distributing (spreading) process, and how this concept fundamentally transforms the statistical modelling.
- Chapter 5 proposes an answer to the questions from chapter 1, by showing that the VF mapping paradigm connects to a more general concept from statistical mechanics. It is explained how the proportional dependencies that interlink the voice production parameters reflect on the integration from microscale to macroscale. The balloon-in-the-box is chosen as an illustration to explain why the exponential factors that reflect in the individual quality parameter distribution shapes (and that control the individual bounds of the voice range) are not comparable between individuals in an absolute sense, while still being specific and reproducible within a subject.
- Chapter 6 presents a single case study to demonstrate the specificity that can be reached in describing individual voice change, and to demonstrate the use of several complementary voice parameters. It explains the hierarchy, and in what sense the jitter parameters is a ‘holy grail.’ It is emphasized that the interpretation of a voice parameter distribution is always contextual. It shows in a practical way the sense and nonsense of averaging and the use of norms.

1 INTRODUCTION

1.1 The mapping paradigm

The central theme of this thesis is the mapping of voice quality metrics as a function of fundamental frequency (f_o) and sound pressure level (SPL). This coordinate system is shared with the voice range profile (VRP) and the speech range profile (SRP); however, determining the range of the voice is not the primary interest of the present work. Naturally, it helps to have some prior understanding of what the VRP and SRP are about, and therefore paper **F** is attached, but this is mostly for the convenience of the reader. Paper **F** efficiently introduces these concepts, and it could have served as an introduction to this thesis, had I wished to present it from a more conventional perspective. But my goal here is different – and this also explains why I was not the first author of Paper **F**. Getting to the actual essence of the mapping paradigm requires a different approach.

To guarantee a fresh start, the terms VRP, SRP, and phonetogram are herewith quarantined; they are too closely connoted with testing for bounds of the range, with norms, with protocols and questions on how to reduce. As a placeholder, the term ‘voice field’, a translation of the German term ‘Stimmfeld’, is adopted instead (see Figure 1).

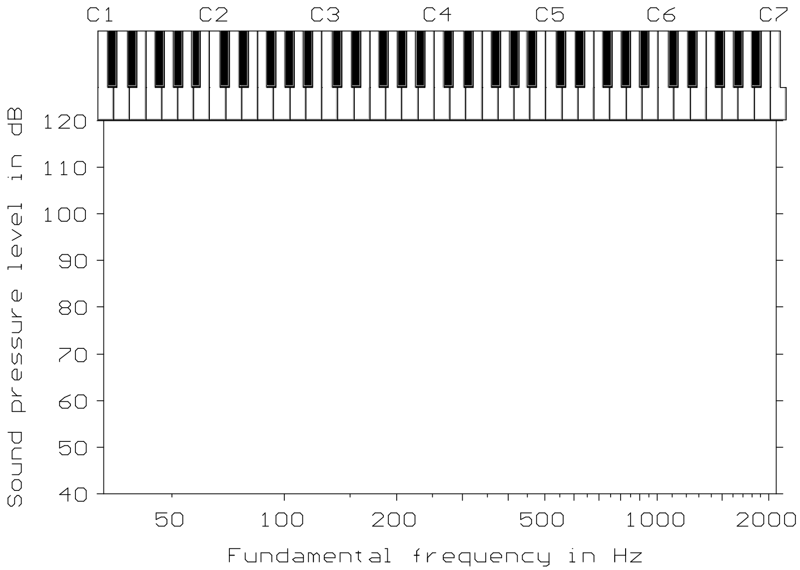

Figure 1. Empty voice field frame.

The term 'voice field' (VF), more accurately than 'VRP', refers only to the area that carries the content, while leaving open what this content will be. Thus, the mapping paradigm does not automatically imply a search for a bound, although we may safely assume that bounds exist. The essence is that the VF can house content by being context free. So, the term 'voice field' is generic, it refers to an open area onto which may be projected any type of 2-dimensional (2D) conditioned distribution, with f_o and SPL as the x and y coordinates.

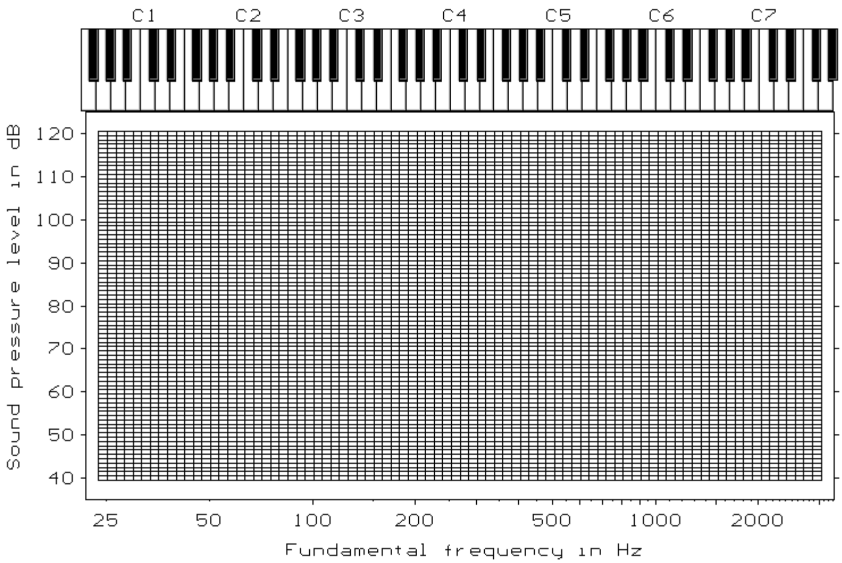

Figure 2. Empty voice field cell matrix.

Additionally, the term voice field matrix (VFM) is used to denote the data storage field, a matrix of cells that holds a sampled parameterized representation of a specific voice aspect (see Figure 2). A cell may hold several parameter layers that each relate to a distinct quality aspect of the voice, or may hold even complete vectors (like spectra) that represent a compound aspect. To give a glimpse of such a complex assembly, paper **E**, (the appendix) contains two high resolution vector PDFs where the reader can actually zoom in on a matrix of cell spectra.

The VF, when represented as a VFM, may house all sorts of measurements or results from different test protocols, or actually their distributions, it is the 'measurement paradigm', referred to in the title of this thesis. It is not just a straightforward mapping of the geography of the voice, it is also about the interrelationships, the mutual proportional dependencies, the 'social geography' of the parameters. Moreover, the sampling of voice quality in metrics is a highly non-trivial issue, for which the rationale is given in Chapter 2. Note that in paper **E**, the term 'parameter' is replaced by 'metric', a term that directly refers to a numerical result connected to distinct detection scheme. In theory, there could be more metrics, or measurement schemes that all try to characterize the same "voice parameter". When explicitly dealing with an actual measurement scheme, 'metric' will be the preferred term that is used in this framework text, otherwise it will be just 'parameter'.

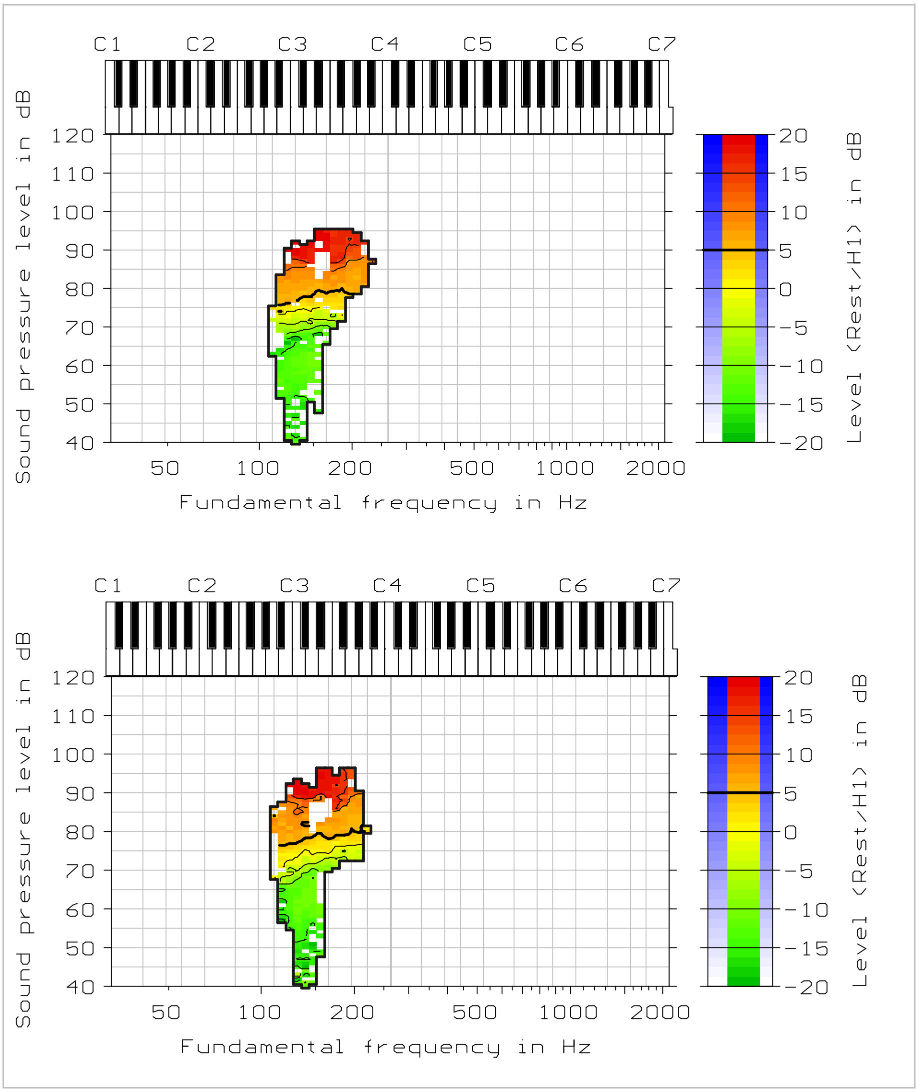

Figure 3. Traces of phonations at approximately the same VF area, each recorded at two different time instances, both using the vowel /a/ only. The voice metric is arbitrary chosen and the colour pattern will not be explained or interpreted at this stage.

The two VF panels in Figure 3 show the results of recordings made at different times. Note how at different moments in time the recorded metric values are largely comparable when the f_o and SPL coordinates are the same. Seen as function of both f_o and SPL, the metric values change in a consistent manner.

1.2 The localizing principle

The VF coordinates are to a large extent prescriptive for the sampled acoustic parameter values. An intentional change in voice quality is generally matched by a concurrent change in VF position. Conversely, a change in VF position generally coincides with a change in metric values. Thus, navigation over f_o and SPL becomes synonymous with navigating to a distinct quality setting. This empirical finding gives the VF mapping paradigm an instrumental property.

The intimate link between VF coordinates and voice quality values defines the practical use of the VF mapping. The distribution shape that is revealed on the map makes it possible to assess small *local* changes in individual voice quality that are linked to a distinct and reproducible voice production setting. If such a voice setting characterizes a distinct singing technique or voice problem, the measured metric values will thus be specifically linked to an addressable position on the map that will act as a preset for a physiological context. By navigating on the VF map, it is possible to later return to the exact address of this voice setting to see if something specific has changed, in relation to this context.

This localizing principle should be seen in the light of the great physiologic importance of the VRP contours, that are personal, and reproducible. The same mechanism that localizes the contours, localizes the measured metric values.

Interactive navigation over the VF map is a key element of the paradigm. This navigation involves real-time detection, storage and display of all voice metrics in relation to the momentary f_o/SPL position in the VF. It is understandable that serious researchers might consider the real-time visual feedback a distracting, complicating gimmick. From that standpoint, though, they run the risk of missing the immediate experience of the strong organising role that the SPL and f_o have on the voice quality. The two VF coordinates are typically more prescriptive for a certain quality than would be expected on the basis of the variance that is generally experienced along the time-line. Ask a person to sustain a tone twice, and the two phonations will never be precisely the same, but when the person is interactively guided to the same VF position, the match can be optimized.

1.3 Practical exploitation of the localizing principle

The localizing property associated with the VF attains an extra dimension when this aspect is experienced *live*, at the moment that direct visual feedback is used during a VRP recording. The voice suddenly reveals itself as a highly *deterministic system*, due to the instant confirmation that the quality parameter is part of a systematic pattern in relation to the absolute position in the VF. This impression is strengthened when the colouring actually varies in direct proportion to the perceived quality. This leads to an integrated audio-visual-motoric percept that comprises a representation on three acoustic scales, that of the f_o, the SPL, and the quality metric, where each scale is connected to a different perceptual feature. This amalgamation to an overall percept presents a sense of *causality*, while at same the time, it imbues a sense of *reliability* connected to (a) the metric, (b) the paradigm, and (c) the system that implements it. This aspect pays the bill, because the illusion is complete and convincing, while it is notoriously hard to reach and maintain such a closeness for a single metric, and even more so for a constellation of metrics. Moreover, when such a level of abstraction is actually achieved, it gives a distinct feeling of satisfaction to see science in action.

From the salesman point of view, the main trick is not to spoil the impression by serving metrics that behave inexplicably in relation to what we perceive. The risk of getting trapped in such a conundrum is high. Any voice quality metric is essentially abstract. Most metrics show confusing patterns, at least initially. The common expectation of a naïve user of the system is that an acoustic voice quality parameter will be directly connected to the perceptual aspect with which it was associated in the literature. Even scientists share the same expectations, based on their many extensive offline sequestered data analyses. We all like to simplify and to make abstractions. However, practice shows that most of the parameter detection schemes published in the literature cannot be copied one-to-one with an immediately fit to the paradigm, nor do they guarantee the desired transparency or responsiveness.

These reservations on applicability have nothing to do with the quality of the earlier scientific research. High-quality research defines the launch pad, it gives the head start. This study would not be possible without this fundamental base. Naturally, I am hugely indebted to the wealth of prior work by others.

The point is that, when measuring on logarithmic scales, the *proportionality* of a parameter becomes the central interest. For an established parameter, the stakes change if this proportionality needs to be evaluated in relation to the *large proportional changes* that the mapping scales prescribe. There are many unanticipated factors, such as normalizations and noise sources, that are revealed only when the question on proportionality is asked. It is hard to convey the craftsmanship that is connected with the identification and control of these factors. This framework text can only sketch the rationale and the mental model associated with the role of proportionalities in describing a voice.

1.4 A liberal principle

Curiously, the above reservations do not imply that metrics show *undependable* patterns. Surprisingly, patterns tend to be reproducible. The localizing feature that determines the reproducibility in the VF mapping is a liberal principle. The VF navigation always guarantees a closest match to an earlier produced acoustic output, actually a signal generated using the most comparable physiologic setting. As long as the detection scheme for a newly developed voice metric does not change, the same (possibly puzzling) colour pattern will emerge. Thus:

Any bizarre metric or erroneous detection scheme may present a distribution shape that will still look trustworthy due to the reproducibility that it attained by the localisation in the VF.

> With modern software development tools, anyone can create, in a limited amount of time, a rudimentary automatic recording system that does VF mapping, and post the application on the internet. Neither the developer, nor the subsequent user of such an application will get the impression that the displayed metric shows a suspicious result, as generally there is no preconceived notion of what the result should look like. The localizing property will work equally well with an uncalibrated vertical SPL scale, and reproducible results will promptly emerge in the VF due to the localizing paradigm. Because there will always be settings where the displayed proportionality will make some sense, the reproducibility could generate a strong, but false sense of relevance and correctness of the implemented metric. Calibrating with a synthetic source that affords an exclusive control of selected acoustic features (and that permits a direct perceptual check,) is therefore the designated way to judge the relevance of any new metric.

Perhaps the most misleading assumption is that the more irregular or strange the observed pattern is, the less trustworthy it is. This reasoning is understandable, but dangerous. It is better to reverse the reasoning:

> Because it is so difficult to go against the localizing principle, the appearance of irregularity (or entropy) can be considered the strongest marker of independency of a metric. Smooth VF-maps are suspicious! These are generally a sign that the observed variation is (A) controlled by the recording system and not reflecting an aspect of the voice, or (B) that the metric lacks normalization and reflects an inclination that is inbuilt by the detection scheme or (C) reflects a trivial voice aspect and not a targeted detail.

There is a very peculiar statistical rationale linked to the VF mapping paradigm. This rationale is the central theme in Chapter 4.

It is important to mention the very positive side of this liberal principle: with no preconceptions on what distribution shapes should emerge, the element of surprise is not pruned. It is difficult to divest oneself of preconceptions; they act as a mainstream ‘force’ that connects all patterns and that always defines the voice production context to which the VF-position refers. New discoveries can be made by searching for metrics that are able to move with this force but still show a new independent behaviour that is nontrivial.

1.5 Assessing reproducibility, how prescriptive is the VF framing?

For assessing the reproducibility that is associated with the localizing principle, it would seem logical to compute the in-cell variance and then to display this objective measure in the same manner as voice metrics are mapped over the VF. However, we argue that this method would be misleading. For instance, the recording system is standardly set to accumulate samples, to do in-cell averaging. However, there is no option implemented in software to show an in-cell standard deviation, and this is by design, as it would go against the fundamental principles on which the VF-mapping paradigm relies. As stated before, there is a totally different statistical rationale linked to the VF mapping paradigm, as discussed in the following (and in Chapter 4).

The recording system is designed such that metrics are preconditioned and optimized to be instantaneously displayed at their target. Furthermore, metrics are chosen to reflect voice quality aspects that are also instantaneously heard. Averaging more voice samples to the same VF storage cell will not imply that the obtained voice quality estimate improves. If during an assessment, the metric value ‘flickers’ at the location of a cell, then there is something wrong in the assessment scheme. There may be some spreading observed in relation to the values for neighbouring cells. This spreading would show as a speckling of the image, but such spreading does not imply an error in the measurement! This is a consequence of the VF-measurement paradigm, where logarithmic scale mapping is a central property. The choice not to display a standard deviation reflects the fact that the mean and standard deviation receive a totally different interpretation when these statistics are to be computed on logarithmic scales. So, let us temporarily ignore the notion of in-cell variance. The VF mapping paradigm is connected to a proportionally distributing (spreading) process, which is a nonstandard statistical concept.

With conventional VRP *contour* recording the *standard statistical reasoning* applies. More hits for a cell may substantiate the finding of a stop or bound. Accumulating more voice samples will help to precisely define the shape of the SRP distribution. However, for the individual voice quality assessment, in-cell averaging is not essential, nor useful, and sometimes even counterproductive. Remember that this thesis focusses on mapping voice quality metrics in the VF and not on assessing the bounds of the range.

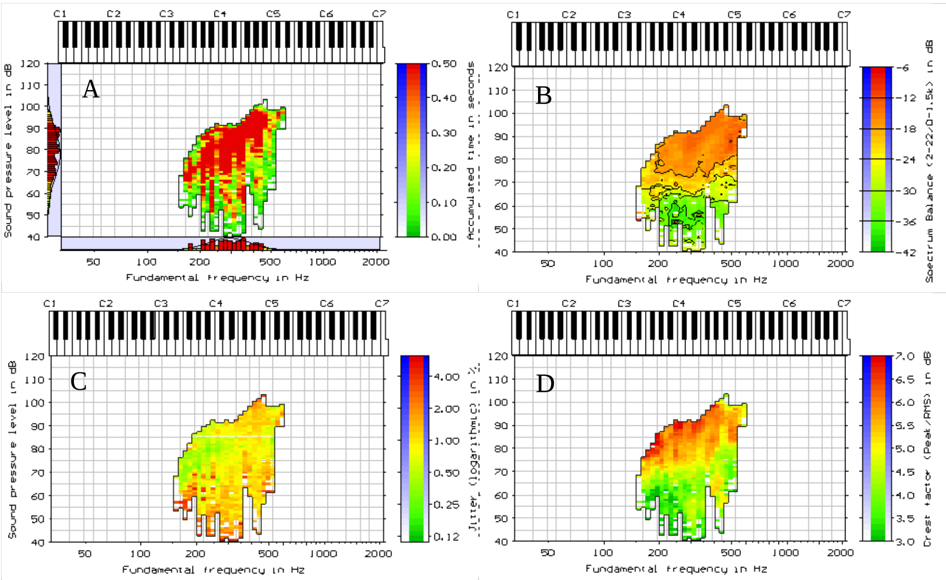

Figure 4. Feature maps for four different metrics. A: Phonation density (accumulated time), B: Spectrum Balance, C: Jitter, D: Crest factor. Source: subject TF1 in the database supplied with paper V.

Figure 4-A shows a map of the ‘phonation density’ metric. This metric has a separate status, as it is not a voice quality parameter; it only counts the hits, or

accumulated time per cell. A red colour marks the cells with a high density or high accumulated time count. The vertical red stripes are a result of swell tones (from soft to loud) that the singer made at these distinct pitches. Note that the cells between the red stripes are covered only a few times (green colour). Figure 4-B maps the distribution shape of the SB metric. The green-yellow-red gradient has a very different interpretation with the SB metric (for an explanation see paper **E**). For the moment, we consider only the abstract colour pattern. Note that the sharp vertically ordered density contrasts seen in Figure 4-A make no impression at all on the SB metric distribution seen in Figure 4-B. The two patterns show no obvious visual relationship. The target values for the SB metric are unknown, and the phonation density incurs no expectation for the distribution shape of SB or indeed for *any* metric. Thus, there is no direct criterion for objectively assessing whether an in-cell estimate improves when more samples are accumulated. As our visual pattern recognition system is extremely good at spotting any associations, this subjective check is by no means a weak option.

This criterion—that the density pattern should not shine through in the metric patterns—is an overall design criterion. All metrics are instead optimized for an immediately responsive visual feedback. An important goal of the VF mapping is to create a map where all time dependencies are eliminated, including the dependency on the averaging time.

For some metrics, independence of the accumulation time cannot be completely achieved. For instance, the jitter and shimmer metrics are of a stochastic nature themselves, and will inevitably show a smoother, less speckled image as the exposure time increases. A detailed inspection of Figure 4-C reveals that in the f_o range from 200 to 300 Hz there are vertical stripes running upward from 50 dB to 80 dB SPL, where the colouring is inversed in relation to the phonation density pattern seen in Figure 4-A. Inevitably, with more jitter observations, the probability of obtaining a high jitter value decreases.

1.5.1 Timeless representation

In principle, the voice quality maps are timeless representations. In this manner, maps from different voice recordings can be compared with little concern for differences in exposure time. It also implies that the spatial distribution can directly be interpreted as always reflecting a quality aspect of the voice and not a peculiarity of the measurement system. Thus, we can step up in abstraction level, as it is no longer *a single value* that characterizes a quality aspect of the voice, but *a spatial distribution* that determines the connected information content.

1.5.2 The distribution only

The thought that the distribution is the leading factor in the description of the voice quality could be taken to the extreme. We are free to ignore the numerical values of any voice quality metric and investigate the spatial distribution only. A patient (or passionate) reader could be challenged to do a purely visual inspection of the feature maps for the individual voices that are supplied with paper **E**. That reader would probably notice that with the trained female voices the spectral variation over the range appears more consistent; the images are less speckled or irregular than for the untrained female voices. The trained voices show unswerving maps, which could imply that these voices are more consistent in their dynamic behaviour. This again would be the effect of training and would reflect a better control or anticipation of the dynamic changes that the voice traverses as a function of f_o and SPL. Of course, the idea that voice training leads to a better control is not a new one. It follows, for instance, the ideas of Collier et al. who tried to objectify it by measuring the SB/SPL [1]. Note that a consistent distribution shape could similarly reflect a loss in alterability, a loss in degrees of freedom due to a dysfunction. It is clear that the consistency of the spatial distribution looks promising as a measure of voice function, and needs more systematic research.

1.5.3 Randomness is not noise

It is important to note that spatial irregularity in the VF is not an effect that is induced by a drowning in noise at low SPL. Noise is a consistent property of the voice. Random deviations are equally probable at any horizontal or vertical position in the VF. What if this spreading in the metric values is actually the result of a dispositioning in the VF, a result of an uncertainty in the f_o and SPL coordinates, a dispositioning error that varies as a function of SPL?

Would there be for instance a slow wavering in level at low SPL, then such a modulation will be traced over time and samples would accumulate over a chain of different cells that cover a vertical band in the VF. Still, over this band the metrics will remain linked to the absolute SPL in a very consistent manner. Fast irregular changes in f_o and SPL are externalised and traced as separate metrics (the underlying concept is explained in more detail in the next section). The period-by-period irregularities in sound pressure (shimmer) or duration (jitter) generally increase with decreasing SPL. However, the real enigma is that the dependency of jitter and shimmer on the absolute f_o and SPL does not waver. The same is true for the spectral metrics reported on in paper **E**. The binding to the VF space is remarkably constant.

1.6 Certifying reproducibility

The localizing principle is a key property of the mapping paradigm, but it is an emergent property, and as explained in the previous paragraph, there is no direct measure for assessing its accuracy. However, reproducibility can be improved upon by identifying the sources of variation that surround the voice quality sampling process. To this end, a synthetic voice source to test and calibrate the analysis is indispensable. The author spent years improving the reliability and reproducibility of the recording system. The following list summarizes some of the practical know-how accrued.

How to improve reproducibility:

- Use the same vowel, typically the vowel /a/.

(Note that according to Lamesch [2], in M2 (falsetto/head voice) the vowel has little influence on the maximum SPL. The effect of formant resonance on the maximum SPL is also questioned in paper **E**. If the same indifference would apply for any SPL in the range, the dispositioning effect of the vowel on the localizing of quality metrics could be small. This effect needs a proper systematic study. The strength and weakness of this thesis is that all results presented refer to a relentless use of the vowel /a/ only.)

- Stay within the same register mechanism.

This strategy is stringently applied for the procedure used in paper **E**. Conventionally, the VRP contour maps the combined areas of the two registers. Direct visual feedback (VF navigation) and an adapted instruction protocol are essential for achieving a differentiation between register mechanisms (see appendix S1 in the supplementary files with paper **E**).

- Preserve a maximum signal to noise ratio (SNR) down to the threshold of phonation.

The full dynamic range of the voice, up to 120 dB SPL, cannot be covered with just a single 16-bit converter. If such a converter is used, all metrics will suffer from a reduced reproducibility at low SPL due to a poor SNR. The problem is increased by the standard 30 cm mouth-to-microphone distance that is prescribed as standard for VRP recording [3]. The SNR increases significantly when a microphone close to the mouth is used [4][5][6][7][8][9][10], but in this situation, it will be difficult to keep the level calibrated. In the VP recording system the problems surrounding the SPL measurement were solved by using a dual headset with two microphones: a close microphone and a far microphone at 30 cm that acts as the reference (see Figure 5). The arrival time differences and level differences from close and a far microphone signals are continuously monitored by the recording system. From the measured time and level differences, the system determines the location of the sound source and also decides if the current calibration changed due to a repositioning of the microphones [11]. A separate ‘watchdog module’ controls the calibration, and also checks for distortion, monitors the current signal to environmental noise ratio, and sounds an alarm if a constant 50 Hz or 60 Hz interference is accidentally picked up from a power line. Regretfully, a more extended description of this system cannot be given within the scope of this thesis. Note that the decision to develop this dual microphone system was prompted by the necessity to have reliable and reproducible quality maps. The dual-microphone setup also secures the threshold level detection that is needed to measure the VRP contour. I am greatly indebted to Trine Pritz, who spent a lot of effort in assessing the quality of this setup for phonation threshold detection [12][13][14].

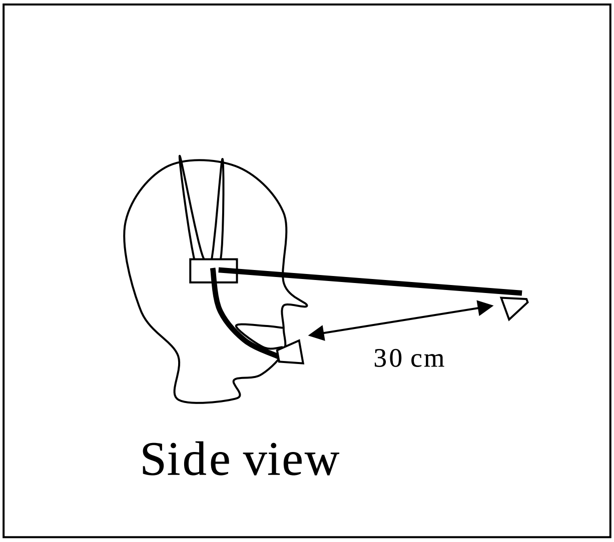

Figure 5. Sketch of the dual microphone headset of Voice Profiler, copied from the user manual [11].

- <u>Guarantee a reliable absolute SPL reference over the full range.</u>

Theoretically, this is a trivial and solved problem, but in practice it is a non-trivial issue to guarantee and maintain a reliable absolute SPL reference. There are many caveats, where protocols need to be followed with rigor. The practice is thoroughly explained by Švec & Granqvist [15][16]. The main difficulty is to *maintain* a calibrated absolute SPL scale. With a fixed microphone sensitivity in dB/Pascal, this becomes a problem of maintaining a constant voltage reference with the conversion to the digital domain. This problem was solved in the VP recording system by adding a reference voltage clipping circuit in the hardware of the amplifier. Using mixed-in signals coming from a private (hidden) synthesis channel that exclusively enter this clipping circuit, the VP software regularly checks the hardware conversion voltages reference markers.

- Proclaim compliance to a pervasive logarithmic scale mapping.

The numerical value of an acoustic voice metric can be an amplitude, an energy, a power, a fraction, a variance, or a magnitude, as long as the number is always positive and never touches scale value zero. This is a mathematical necessity for a mapping on a logarithmic scale. If the metric values were mapped on a linear scale, there would be a conceptual flaw with the paradigm. With logarithmic coordinates, a constant step in the X or Y direction on the map would reflect a proportionally larger (or smaller) magnitude change in relation to the linear-scaled metric. Logically, the graph would become reproducible and even predictable, because the observed dependency would actually reflect the change in resolution in connection to this dissimilar linear scale. At the same time, the range over which the linear mapped metric would show a contrasting effect would collapse. Thus, we would see little effect of the voice, and a lot of the system. So, *a logarithmic scaling is mandated on all axes.* Thus, a reproducible pattern will acknowledge this particular property of the voice, that the proportional change in a voice quality metric depends totally on the absolute magnitude of the sound pressure in Pascal (the vertical coordinate), and the absolute frequency in Hz (the vertical scale). This last property is the basis for the localizing principle.

- Model on exponential scales.

To comply to the log-scale environment, a metric needs to be modelled as a signal that changes exponentially with time. For the implementation, this means that all metric 'macroscale signals' are up-sampled on a micro-timescale (the sampling clock rate), using exponential interpolators. Thus, running mean and variance signals are derived that are actually running geometric mean and running geometric variance signals (or geometric standard deviation signals, if the exponents are halved). Although the period duration is treated as being constant over its own linear time frame, the geometric mean period duration and geometric deviation signals (the jitter, frequency vibrato and wow) are still modelled (and filtered) as exponential dependencies on linear time. These assertions will probably be puzzling to the reader. The underlying statistical reasoning that is connected with the concept of geometric means and deviations is discussed in detail in Chapter 4.

- Synchronize the sampling of all voice metrics, including the SPL, to the fundamental period.

The VF is a timeless representation where all mapped metrics are bound together by a set of proportional dependencies. The constellation of metric values that all share the same cell coordinate is considered one coherent ensemble, a meta-sample. This entity (meta-sample) was sampled at one particular moment in time. A specific proportional dependency between all metrics will exist at any instant in time, but the temporal evaluation of this dependency is not the issue here. The phonatory period 'clock' determines the rate at which this coherent unit is sampled at the timeline, and also the duration over which the information in this coherent unit is valid. Note, that in paper **E**, all spectral metrics are derived from a harmonically sampled spectrum envelope, and thus all spectral metrics take the duration of one period as their common base. Thus, the same period-synchronized concept is implemented. When the f_0 varies, the validity period, the lifetime, or 'information bandwidth' of the meta-sample changes accordingly. This 'information bandwidth' is again proportionally the same for all metrics in the ensemble, but the amount of averaging time that is seen in the perspective of a single metric, varies between metrics. But let us stop here. Time sampling of voice metrics is a nontrivial issue, where particulars in the implementation may exhibit large effects. A proper sampling model clears a lot of mist and significantly improves the localizing. Moreover, the perceptual relevance improves, as typically perception, too, implements proportional scaling. For more details, see Chapters 3 & 4.

- Use optimized, cleaned, independent parameters

These remain vague terms as the optimization and cleaning is such a messy process. It is a strange mix between voice science and engineering. There is no model that is able to predict what the distribution shape for a metric should look like. Interactive testing with a parametric synthesizer remains an essential starting point (as explained in Chapter 2). Experience is important. Typically, new metrics are defined and evaluated in relation to metrics that can be relied on and that can act as a reference.

The perturbation metrics (jitter and shimmer) play a special role. Both perturbation metrics assess a signal-to-noise ratio (SNR) that ideally should reflect on an irregular vocal fold vibration. However, the same metrics are sensitive also to the presence of external noise sources that influence the measured aperiodicity. Moreover, jitter and shimmer metrics present an undisturbed view on the voice source only, and ideally show a distribution shape in the VF that is not affected by the vowel.

> The shimmer metric is implemented in the VP recording system, but it is not advertised as a voice quality metric, because it is such a sensitive metric for revealing external noise sources. The shimmer,

> i.e., the short-term average period-to-period peak amplitude deviation, generally changes with the mouth-microphone distance. This is because the perturbing effect that reflected pressure waves have on the period peak amplitude will grow when the amplitude of the direct sound component decreases. Within a more reverberant recoding environment, a switch from a close to a far microphone generally shows as sudden change in shimmer, given that the metric is so sensitive and can resolve this effect. If a recording system does not show this sensitivity for shimmer with a reverberant room, then its implementation may be called into question. A gradual decrease in period-to-period perturbation with increasing SPL is a natural property of the voice (as is discussed in the attached papers). Curiously, shimmer values may very suddenly decrease when the voice becomes very loud. Such a sudden decrease may be the effect of a soft clipping due to a saturation in the microphone that soft-clips the period peak amplitudes such that the peak-to-peak variation is reduced. A digital (hard) clipping of peak amplitude levels leaves an even clearer mark on the shimmer metric. One of the implemented jitter metrics measures the short-term average period-to-period duration deviation, with the period peak instances as the duration markers. Such a jitter metric may reflect, in a convoluted way, on peak clipping too. In the Voice Profiler system, a set of different jitter metrics are implemented that each trace different period duration markers, and thus respond with a different sensitivity to period-to-period fluctuations. It will be evident that there are many pragmatic and unsubstantiated choices made when a metric is to be selected to be sold as 'jitter parameter'. There is generally no ideal or best metric.

This last rule is true for any voice metric. A metric is better within the perspective presented by an ensemble of voice metrics, where none of the metrics has an absolute scale mark that can be used as a golden standard. The interpretation of a VF distribution shape for any metric is inevitably contextual. The context changes with the voice register, so a jitter metric will have a different interpretation in M2 than in M1 (see Chapter 6). There is a hierarchy in metrics, but even this hierarchy is not absolute, as it too changes with the context.

1.6.1 Interpretation

There is generally no ideal or best metric, but there is a best metric given the context. There is no best metric for all voices in general. The optimal metric to contrast the qualities within a single voice takes into account all the observed

acoustical variation that this specific voice is able to display. If the full f_o/SPL range is scanned all possibilities are essentially known. This approach is more or less the starting point in the EGG metric mapping experiments of Selamtzis & Ternström [17][18][19]. Note that purposely aiming *not to be general* but to observe what happens with each individual can be a very good starting point to find the best generalisation later.
Within a voice, a distribution shape makes most sense. When averaged across several voices, it only becomes more difficult to interpret the results. Averaging across voices is seldom helpful. If all voice metrics would refer to the same expected value, then averaging would be helpful, but here, the reasoning is different. The spreading within a single voice recording is not random, and the deviations within a mean VF-distribution shape are not random 'errors'. In the log-scale VF-mapping paradigm the concept of variance (standard deviation or spreading) has a different interpretation. It acquired this interpretation already during the assessment (see Chapter 4). This interpretation will not automatically change when metric distribution shapes are averaged between or within subjects.

Let us consider an example. A jitter metric effectively reports on a standard deviation (the square root of a variance) linked to a specific VF cell. With in-cell averaging across voices, the new jitter value for the cell should logically be calculated by averaging variances, where the 'N', the total number of accumulated samples will increase, and where the mean to which the new variance refers is also updated. However, the two variances that the jitter values from different voices represent do not necessarily refer to a deviation from the same expected value. Remember that the information in the VF cells is no longer connected over time, and that metrics are designed to be instantaneous at their target, where thus the N for a cell should not matter. The variance observed for a voice, which is quantified by the jitter metric in that cell, has a scaling factor that is typical for that voice in that cell: the localization principle works for the jitter, too! The expected value of the variance is differently scaled between voices. When logarithmically-scaled jitter values from different individuals are averaged per cell, then these personal *scaling factors for the variance* are averaged, which is a fundamentally different concept. Moreover, the logarithmically scaled jitter was sampled from the time line as part of an ensemble of metrics. Multiple metrics were sampled at the same moment in time, as one 'meta-sample'. The domain from which they were sampled was modelled as an exponentially varying magnitude domain to ensure that all variances and standard deviations became geometric variances and geometric standard deviations. Thus, all voice metrics, including the ones that describe the variance, maintain their *mutual proportional connection* from the first instance that they were observed together.

The concept of a geometric standard deviation (variance) is further elaborated upon in Chapter 4.

1.6.2 Qualitative and quantitative are not mutually exclusive

With different personal distribution shapes, it is difficult to make generalizations, but that does not mean that the assessed results are vague, imprecise or not reproducible. The typical (and perhaps only viable) approach in interpreting distribution shapes is a pragmatic one. It is better always to consider an assessed VF distribution shape as the result of a goal oriented test that shows *a specific* quality setting for *a specific* voice in *a specific* singing task, where the hope of ever finding a generalization is disregarded from the beginning. Metric distributions are necessary contextual, shapes are comparable, but not in an absolute sense. Thus, any analysis or interpretation typically starts from a qualitative inspection of characteristic shape aspects. As explained in paper **F**, a VRP can be considered a collection of tests, so it is important that the original goal of the assessment is already known with the interpretation.

Note, that a *qualitative* judgement does not preclude being *quantitative*. All metrics are localized and numerical values remain hard and significant within their context.

> In connection with the selling of the Voice Profiler recording system (that reached a peak around 2006), a general instruction course was developed and offered to the users. Later, an advanced user course was developed on how to interpret the metric distribution shapes (and how to use the metrics for direct visual feedback). For this last course, a ‘Top-6’ interpretation scheme was defined [23]. The scheme prescribed a zooming in on several voice-features following an initial qualitative evaluation. The overall approach was criticized for being qualitative, as in “not scientific,” which would have been true if there were not a solid quantitative base underneath. Later, this Top-6 scheme (but not the connected vison) was shamelessly copied by the competition.

1.7 Spotting the nonlinearity

The feature maps with the metric distribution shapes typically characterize nonlinear dependencies, that on a logarithmic scale appear as linear dependencies. The term ‘linear’ typically refers to a linear scale model, where the qualifier refers to a linear (invertible) dependency between the input and output of a system. The qualifier ‘nonlinear’ has a different interpretation and materialization on the logarithmic scale. The sampled logarithmic scale represents a different continuum.

In the feature maps presented for the spectral metrics in paper **E**, repeatedly a +14 dB/octave reference line is drawn. Figure 6 and its caption, that were once part of the manuscript of Paper **E** (before it was revised), shows the origin of this reference line. This pink line that survived the subsequent revisions, sketches the constant slope of the VRP contour (the maximum SPL as a function of f_o) that on the average is observed with female voices phonating in M2.

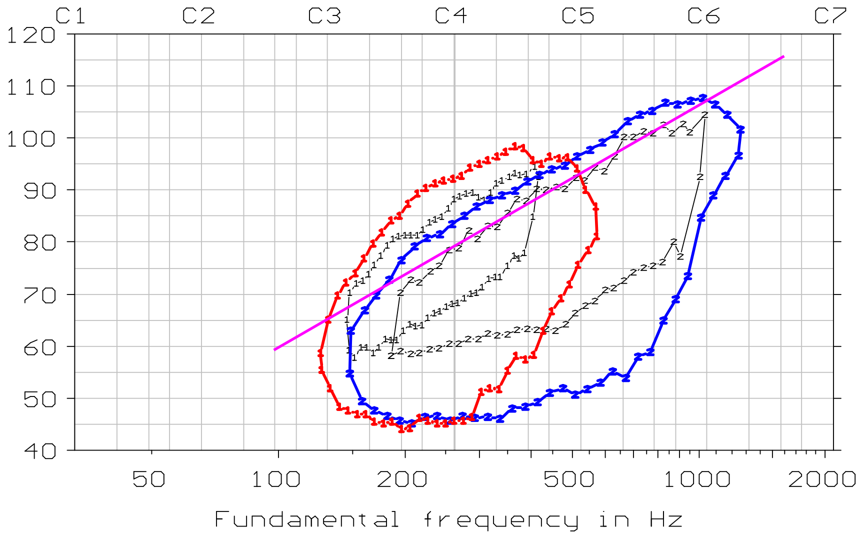

*Figure 6. Mechanism-specific average VRPs for the group of female singing students (TF) from Paper **E**. M1/modal is shown in red and M2/falsetto/head in blue. These averages are juxtaposed against the average contours for mechanism 1 & 2 (both in black) from Roubeau et al. [24] for female singers. The black curves were shifted +7 dB to compensate for the larger microphone distance used in that study (100 cm instead of 30 cm). The pink straight line at +14 dB/octave is given as a visual reference.*

Engineers who try to interpret the VRP contour curvature or metric patterns, sometimes expect the sloping in the VF to discretise into multiples of +6 dB/octave. With this expectancy, they display a conceptual unawareness. These are concepts from a microscale that are falsely assumed to automatically ripple through to the macroscale (this conceptual distinction is explained in Chapter 5). The 6 dB/octave is a numerical mantra from filter theory. It links to the lumping of discrete electronic parts or discrete feedback delay units, to discrete derivatives and discrete differentials. The integral voice production system is continuous and is not bound to these constraints. Derivatives and differentials can be modelled as being fractal [25][26][27]. The expected discretization in fixed slopes reveals only a constraint in the conceptual world of the engineer. Derivatives and differentials are modelling concepts that refer to micro-time or micro-space dependencies that always average out. The metrics that are mapped in the VF-mapping paradigm can only show macroscale properties.

1.8 Parameter pragmatism

The following bulleted list recapitulates, in no particular order, a series of statements that can be used as a guideline when parameter distributions are to be interpreted. The practical application is demonstrated in the case study that is presented in Chapter 6.

- Density / accumulated time is not a measure of voice quality.
- Density patterns should not leak through into the metric patterns.
- A voice does not have a single metric quality but shows a distribution of qualities over the VF.
- VF-distribution shapes are comparable, but not in an absolute sense.
- There is no generalised method to extract a characteristic jitter value for a voice.
- There is no location in the VF where all voices are guaranteed to exhibit a minimum jitter.
- Even the most perturbed voices may show very regular vibrations in some regions of their VF.
- Even the most perfect voices may show high jitter values over a large part of their VF.
- Metrics may stay the same, yet the perceptual aspect they represent may change over the VF.
- The same metric variation may be perceptually irrelevant in a different part of the VF.
- There is no best metric in general.
- Metric distributions will sometimes reflect an aspect of the measurement system rather than of the voice.
- Metrics will never have a clearly separable source or filter related interpretation.
- Perceptual interpretations can be specified for metrics, but such links are never wide-ranging.
- Metrics are valid in the instant and not smoothed over time, they are designed to have this property.
- The interpretation of a metric distribution is always contextual.

- The perceptual identity related to a signal property may shift over the VF. For instance, jitter and shimmer sound different at high or low f_o.
- There can be more metrics that reflect on the same aspect, with different sensitivities.
- Linearization, smoothing of distribution shapes (or of VRP contours) is never the goal.
- Smooth distribution shapes are suspicious.
- A further abstraction of the mapped information can be meaningful, but it is wise to have a basic conception of the unconventional statistical thinking that underlies the map.
- All metrics do not have the same relevance for all voice types.
- Parameters can be conditional or compound and reflect a combination of features.
- The whole spectrum can be considered a compound metric.
- Parameters may reflect a state-bound property of the voice.
- An entire distribution shape may change with a change in singing technique. Within such a new context, patterns are again reproducible.
- An entire distribution shape may reflect the result of dysphonia. Voice therapy or treatment may cause complete changes in the distribution shapes for all metrics.

1.9 Conclusion

The papers in this thesis span a long period. Seen over this time line, the recording system was improved, the navigation principle became formalized and more reliable, and the metrics were optimized, leading to more reproducible and sharper images. All these improvements did not lead to a more unified image of how the voice quality changes over the VF. With more supplementary metrics explored, the distribution shapes on the maps just became more diverse. It only became clearer how difficult it is to make generalisations and how difficult it is to compare voices to each other.

Voices differ in many different respects. These individual differences were always there, but were rarely revealed, there was no map to resolve them. The exposure of these individual differences was a prime objective of the present work. Thereby it should be stressed that the logarithmically scaled maps resolve individual differences in proportional dependencies, and not in linear dependencies, which are the received wisdom. When the voice changes in f_o and SPL, the proportional dependencies that are observed in the ensemble of voice parameters always carries a distinct signature, that defines a personal acoustic identity. Many quality aspects are still missed or remain unresolved at this stage, but that does not change the fact that these aspects could eventually be represented in the same framework.

The repeated observation is that the distribution shapes for the different voice metrics are only superficially comparable between voices. This conclusion is supported by the vast amount of data presented in paper **E**. Regretfully, there are not many other studies where metrics are mapped over the VRP that endorse this message [17][18][19][20]. Reaching reproducibility and comparability within subjects seems to imply an overall loss in comparability between subjects. Although this contrasting impression is not an experience that is confirmed by a lot of external sources, it is still considered a central outcome of all the research and technical developments that are reported on in this thesis. The same message could be formulated as: as long as the individual distribution shapes are not resolved, voices cannot be compared mutually.

The two absolute coordinates (f_o and SPL) are no guarantee that acoustic voice parameter values will be comparable in an absolute sense. As long as there is no point on the absolute coordinates of the VF where all voices comply to the same dependency it becomes difficult to define a norm. As mentioned before, normative maps for the different voice metrics are not defined because their creation was not pursued. Norms would become realistic if for instance an external conditioning (such as belonging to the same SATB voice group) would immediately be reflected as a match in distribution shape for the feature maps, or a similarity in VRP contour shape. If such an easy immediate connection had been observed, the research reported in this thesis would already be closed by publishing such normative data. If the causalities in voice production were that plain and simple, there would have been an easy exit.

With a large group of clinical users of the commercial VP-recording system there was an unceasing call for norms. Paper **C** presents the tools to average and compare VRP contours, but avoids calling these averages norms. The necessity to define norms is recognized, but the topic is deliberately avoided in this thesis. It was considered more important to focus on the differences between voices, and to open up a discussion on the origin of these differences. With the presented material, an educated decision can be made to what degree an average can be considered normative. The software tools to create average VRP contours and to average quality metrics over the VF are available to everyone [28][29]. The remarkable diversity in qualities that is revealed and valued in this thesis explains my personal reluctance to do this.

Distribution shapes can be so different that it sometimes appears as if some voices are subject to a completely different mechanism. It is as if there is no common force factor that imposes the same dependencies for all voices. Although patterns can be very different between subjects, seen within a subject the distribution shapes are generally reproducible. So, there must be a common force factor, as all voices comply to the same localizing principle. Here, the SPL is probably the most critical factor in the hard-framing of the personal dependencies. The subglottal pressure can be considered the main force that controls the power of the voice [30][31][32][33]. There are studies that show a linear dependency between SPL and the logarithm of the subglottal pressure [33][34][35][36][37][38][39]. There is little reason to doubt the fact that the subglottal pressure and the acoustic pressure are proportionally related. The logical question in this respect is: Is the subglottal pressure an absolute control parameter too?

What is the exponential factor by which the SPL is linked to the subglottal pressure? Does this factor vary between voices? Does this factor change over the voice range? How much is this factor controlled by the physiology of the individual voice? The subglottal pressure varies only over two decades, while the acoustical pressure that controls the SPL easily varies over more than three decades. The ensemble of spectrum slope metrics that was tested in paper **E** revealed that there are large individual differences in the way the spectrum power increases with SPL. Even within the same voice group, there are large individual differences in the number of decades in sound pressure that can be covered at a given f_o. How is it that, with the same mechanism and same driving force, there are still such large individual differences?

A possible answer to this intriguing question is formulated in chapter 5. The formulated answer is actually a by-product. This thesis was not written with the intention to answer the above questions in a quantitative way. However, it is impossible not to contemplate the above questions if the distribution shapes for the different metrics are to be interpreted (in relation to a hard f_o/SPL reference), as is attempted in the attached papers. To make a useful interpretation, it is good to understand the rationale connected with a statistical analysis model based on logarithmic scales which VF mapping paradigm implements. This is the topic of Chapter 5.

2 HISTORIC CONTEXT AND PERSONAL PERSPECTIVE

2.1 Start of the project

The question of how the quality of the voice is represented in the acoustic waveform and how these qualities could be parameterized and mapped in the VF first became relevant during a research project entitled: “objective recording of voice quality” granted by the Dutch “Praeventiefonds” (1981-1987), under the guidance of Prof. Reinier Plomp. My colleague Gerrit Bloothooft, who pioneered the project, remained connected as advising mentor, a notable role that he maintained for many years even when the project had stopped. The aim of this original study was to record and map a series of clinically relevant acoustic voice parameters over what, at the time, was still called “the phonetogram” the classical name of the voice range profile. Peer reviewed papers **A** and **B** stem from this initial period. It was during this project that the dream was born to build the ultimate voice quality recording system, based on the paradigm of mapping parameters over the Voice Range Profile. At the same time, an ambition arose to be the first to publish the definite VF-map that sketches the general acoustic backbone of the voice. A map that would become the norm. Knowing about this ambition and its historic context is essential for understanding the rationale behind the questions that are considered in this thesis.

2.2 The Dream

> The ultimate voice quality recording system should be capable to record all aspects of individual voice quality in great detail, using hard numbers that are unique for every voice, but still generally applicable. The system should present a comprehensive overview, using simple and distinct metrics, while not giving in to any external pressure to reduce, to simplify or to present a norm. It should measure with such an accuracy that even the smallest change in voice quality can be traced in a reproducible manner. Any measured change should be directly affirmed by a perceived change in voice quality and vice versa. An instant, intimate link between production, perception and visualized parameters should be preserved at all

> costs and define the design of the system. When possible, the real-time synthesis of an isolated acoustic target feature will be the reference to optimize the performance of a metric. Staying as close as possible to real-time is imperative, as any delay could open the door for a detachment from the pursued reality. Only those voice metrics that survive this instant confirmation test stand a chance to be accepted. To become an actual part of this system, every parameter has to be audited again. An analysed parameter [A] needs to strengthen the group, [B] prove its ability to produce a meaningful response over the full pitch and sound level range that the voice is able to cover, and [C] should work with any voice type, no exceptions. Broad scientific acceptance of a parameter will be an advantage, but never a permit. practical application will determine the final standard. If, despite a thorough theoretical basis, a parameter is unable to meet these practical standards, then the theory (the model), needs to be adapted. The targeted application is to record individual voice change, where the system should work as effectively in a clinical setting as in the singing practice. The final goal is to derive a timeless parameterized representation with such an amount of detail and completeness that the original quality of the voice sound can be completely recovered and resynthesized from the stored data, where the original singer and all peculiarities of that specific voice are still recognizable, and even editable.

At the time, the dream was big, but not too big, technically feasible in a foreseeable future, and full of interesting challenges. A dream sometimes turns into a nightmare, and as it is with nightmares, especially these need to be told in order to understand and accept the intrinsic message yourself. So, what was the nightmare?

2.3 The nightmare

The above dream predestined a process of gradual detachment. The pursuit of the dream, getting step by step closer to the goal, also distanced me from the centre of gravity around which acknowledged voice science rotates. Many of the acoustical causalities that the scientific models envisage have to do with formant resonances and an independently acting voice source. In voice science, a voice typically has only one quality, a healthy voice is perfectly periodic, a perfect match in frequency makes you a great singer or gives your voice support or ring. It is a world where standard statistics always applies and where the average defines the norm.

The image is slightly overstated. The general ideas are not wrong, but they are not the only conceivable foundation. There are different ways to interpret the same reality, where acoustical causalities are interpreted from different angles, that work equally well, and may even be better in practice. Are these alternative acoustical causalities different in principle, or do they just appear different due to the perspective? Let me try to be more concrete.

We do not need formants to identify the voice. We do not need formants to identify the vowel. We do not need formants to identify the voice type. We even do not need a specific voice source influence to support these classifications. These are controversial statements, to be sure, but that there is some truth to them is demonstrated by the 'spectral gap demo':

[http://kc.koncon.nl/staff/pabon/SingingVoiceSynthesis/CutoffF3/CutoffFreqF3.htm].

The real strength of this demo is, that it shows that the quality of the voice, even the identity of a voice, can be completely determined by *something that is structurally missing* in the spectrum. This weird 'voice quality aspect' was formulated as an inevitable inference, following a huge (big-data) meta-analysis of the total spectral VRP database that is supplied with paper **E**. The spectral VRP is an all-inclusive representation. For the vowel /a/, all possible spectrum envelope manifestations over the full f_o/SPL range are known, for all participating singers. Thus, it can be decided which spectral bands never carry much energy, in any voice setting. This unconventional angle and its connected reasoning is a direct result of pursuing the dream. The presentation of the spectral gap demo at PEVOC [2015] lead to a further distancing from the centre of the field.

How can one report on the functioning of a voice recording system that tracks the pitch without ever searching for periodicity? It is a great approach, as it works for every voice, in any condition, and it sold a lot of systems without any customer complaining about pitch errors. The method is not easily reported on, the statistical starting points and the line of thinking are so different, but the end result looks the same.
How can one justify theoretically a nonstandard voice metric like the relative rise time, or the crest factor? The crest factor presents the most immediate reflection of the main dimension of voice change over the VF, and thus plays a central role in controlling the tracking in the analysis system. It also plays a prominent role in papers **A**, **B** and **D**, but the theory behind it is meagre. The HRF or $L_{\mathrm{Rest/H1}}$ metric in paper V reflects more or less the same aspect as the crest factor, and has many more references. However, only the last metric will probably find a broad acceptance. I think I understand why the two metrics reflect a comparable

feature, but it is almost impossible to explain using the standard theoretical reasoning. Having a good equivalent, will also imply that no one will ask anymore what the underlying mechanism is that binds the two, while this is such a fascinating subject.

Should the thesis describe also the dual-headset based voice tracking system [11] that relies on cepstral decorrelation techniques and a statistical independency criterion to isolate the voice source from two microphone signals? It is simple if you get it. Everyone can read in a book how to accomplish this. Should I explain the procedure in this thesis because it is a nonstandard technique in the voice community?

How can one properly motivate a recording paradigm that repeatedly reveals how different voices are, but that also reveals how voices differ in aspects that are not suggested by the standard models? In the recording system, conditional voice metrics are implemented. These adapt to the spectral context and have no fixed acoustical template. These metrics have a great practical use, but are difficult to explain to a world that models acoustic voice quality as being controlled by fixed spectral or temporal markers.

What to do with the exponential sampling concept that underlies the VF mapping? How self-referential is this model?

I recognize that I have come to rely on several private conceptual constructions, and with me, so have many of my customers. It helps to have thousands of VRPs recorded and to have analysed many voice sounds in detail using ‘hard’ scales, so I am surely not operating in a niche. The realization that getting closer to the dream also implies not being able to explain my underlying questions, and not being able to refer to a lot of external sources, that is my nightmare and challenge with this thesis.

2.4 The role of synthesis

Within a scientific community that is primarily concerned with voice analysis, voice synthesis is too rarely considered a feasible alternative. By using *synthesis*, many voice researchers could have proven for themselves, in a few minutes, the same finding they have derived with an extensive, well performed and well documented *analysis* of real voice material. In many cases, the confirmed hypothesis, or the effect that was found, is completely controlled by the implementation of the analysis model and the connected sampling. Hence, their findings prove no truth beyond this point, barely presenting new information on the voice, although it is generally believed it is news, because the voice supplied such unique test material.

None of the ca. 200 customers that bought the Voice Profiler recording system ever uses the synthesis option, even though it could demonstrate what the jitter values they just obtained in analysis represent acoustically. When trying to understand what their jitter values actually mean, most customers lose their way in the extensive literature on this subject. More researchers should sonify the data extracted by their models; and many will be shocked by how far from reality they will land.

2.4.1 Start from synthesis

All analysis models are in principle invertible, and the resynthesis procedure may reveal where, and what kind of information is lost in the analysis process. To maintain a constant tracking of the information content that is contained in a voice analysis model, it is thus wise to develop a voice synthesis model in parallel, and even to completely start from synthesis. With each successive development of a next generation VRP recording system, for which results are presented in the successive papers, the integration between analysis and synthesis model increased progressively. Figure 7 shows the scheme presented in one of the two papers [21] presented at the conference 'SMAC 93' (not included in this thesis) and demonstrates the underlying design philosophy.

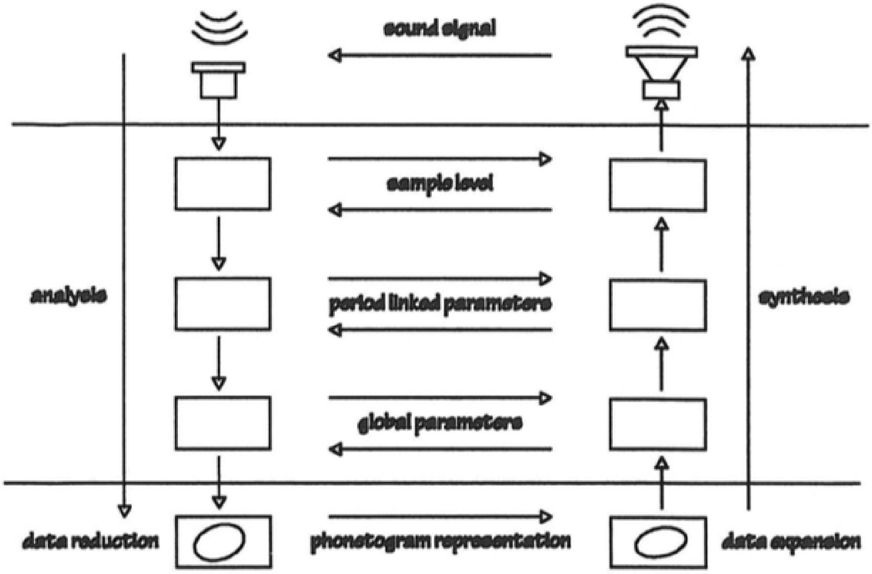

Figure 7. Scheme from the paper: The phonetogram in singing voice analysis and synthesis, proceedings SMAC 1993, Pabon P, 228-23. Original caption: Processing stages in the combined voice analysis/synthesis system. To the left the analysis chain, from microphone signal to phonetogram. To the right the synthesis chain from phonetogram to sound signal. The two vertical lines enclose the DSP part of the processing. Horizontal arrows indicate optional shortcuts between the intermediate data representations.

A combined Synthesis-Analysis system design approach has the following advantages:

1. Modular development & system testing.
2. Identification of information sinks.
3. Routine calibration and remote maintenance.
4. Optimize and "clean" parameters, shortcut the interpretation.
   I intended to measure this… but actually, measured that….
5. Explore and understand sources of co-variation & interdependencies.
6. Interpret analysis results in terms of a synthesis model, preferably a parametric source-filter model.
7. Apply analysis-by-synthesis.

Item 6 reflects the initial ambition formulated in 1993 to sketch the backbone of the voice in terms of a source-filter model.

> The original plan was to derive a source parameterization in terms of an LF-model [22], together with a filter abstraction in terms of formant frequencies, and bandwidths. With the parallel-running source-filter synthesizer as depicted in Figure 7, it was possible to follow the *analysis-by-synthesis* approach. This approach implies the parallel synthesis of a voice signal (right branch Figure 7) that, via an optimization criterion, tries to match the quality of the incoming voice signal (left branch Figure 7). Next, the source and filter settings of the synthesizer are presented as the analysed parameters. With a real-time implementation, the quality of the estimation can be instantly judged from the parallel-synthesized counterpart, which largely speeds-up development time as the type of mismatch can directly be heard. This setup was pursued with vigour, and a vast amount of development time was spent on experimenting and making this idea work. Would it not be great to show a VRP that directly reflects the voice quality in the familiar terms of formant frequencies, bandwidths and source parameters?
>
> To cut a long story short, the approach seemed viable at the left part of the VF (low f_o), but not at very high SPL, neither down to the threshold of phonation, nor at high f_o. The principle worked fairly well with one male voice (a colleague). However, the few tests with female voices and a trained male singer, immediately showed that the principle would never become generally applicable. There were too many cases in which it was problematic to find appropriate (realistic) source and filter parameter settings for the synthesizer to properly match the incoming voice sound.

The above plan was certainly ambitious, and the implementation was correct, as could be instantly confirmed by the reproduced sound quality. Was the modelling principle incorrect? Do voices differ in their acquiescence to a linear separation into source and filter contributions, perhaps due to a nonlinear interaction between the two? This suspicion became stronger and stronger.

What could be the problem with source-filter modelling? As stated in the preface: we understand the voice through our models. Let us take a more distant standpoint and inspect how the actual information on the voice quality can be conceived within a model that equally values analysis and synthesis schemes as carriers of the same knowledge. Figure 8 shows a more recent version of the earlier 1993 scheme that appeared in Figure 7, now augmented with blue arrows that denote the flow of information at the different data-abstraction levels.

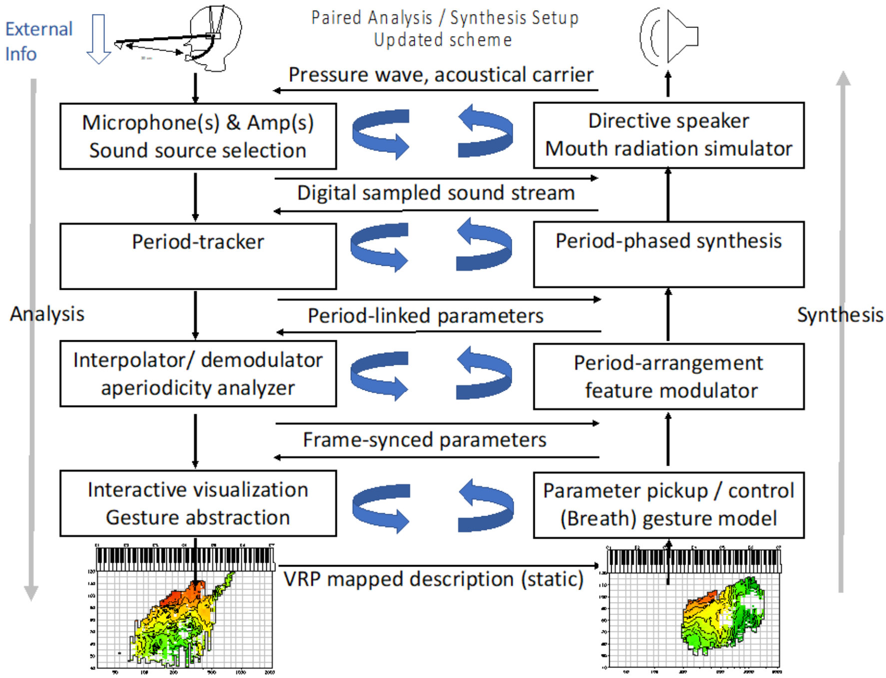

Figure 8. Updated version of the previous SMAC93 figure, blue arrows mark the information loops.

The ability of an analysis tool to characterize the quality of the voice largely depends on the sound material that was used to test and normalize the system. One consequence is that results tend to be cast into the mould of the model that is chosen as a template. Note, that in the scheme of Figure 8, the left top is the only entrance for new, raw information. At all other levels, information is circulating for which the variation in the data is intrinsically formatted or moulded according to the initial synthesis model. A potential danger with the above closed-loop approach is that the system becomes ***self-referential***. Thereby, it risks endorsing our view on the relevance and generality of the modelled parameters, in this case, the preconceived parameters of the source-filter model. Moreover, by templating with the source-filter model, there is a risk of becoming oblivious to important factors that the model does not anticipate, for instance the effects of source-filter interactions. What if these are the factors that actually differentiate and characterize individual voice quality?

The difficulties encountered in trying to implement a parameterization in terms of a linear source-filter model generated a strong motivation not to use any premeditated or preconceived model to interpret the voice change over the VF. It made me decide to store only the 'raw' spectrum over the VF, and use that as a starting point for further research. The downside would be that all mapped voice parameters would be abstract, in the sense that there is no longer a separate source or filter interpretation. This 'open mind' approach eventually led to the set of concepts that is presented in paper **E**.

2.5 The dream coming true

About 20 years, one non-commercial and two commercial "recording systems" later, around 2007, the initial dream of creating the ultimate recording system found a form of completion. With the commercial release of version 5.1 of Voice Profiler Spectral, it became possible for all users of this system, to generate, from the stored VRP parameters, a synthetic voice with a quality that is largely indistinguishable from the original. Let me give an – inevitably positively biased – impression of the level of abstraction:

> Yearly, the VRP recording system is demonstrated for the group of new singing students of the Royal Conservatoire in The Hague, where one of the students is recorded live. At completion of the recording, the resynthesis option is demonstrated. Generally, all members of the audience immediately assume that they are listening to an earlier audio recording of the same voice. The confirmation that they are actually listening to a resynthesis happens when impossible tonal jumps, unnatural vibrato settings, or extreme perturbation values are introduced. The true surprise came, when students were able to recognize, from the unmodified VRP resynthesis, the voice of a colleague who was not present on that day and who had been recorded before in a different setting. For those students who had never heard this person's voice, the synthesized identity remained abstract.

Instead of going into the details of the implementation and the control, and of what possible research questions could be answered with this system, let us consider what this anecdotal observation tells us, and later pass judgment on the actual achievement.

The resynthesis does not use an intricate parametrization. It is based on the unreduced narrow-band spectrum, accumulated per VFM cell, for which the harmonic spectrum envelope is a template[1]. The parameterization is not the miracle. The implication of the recurring episode recounted above is that *voice quality can be uniquely identified by the spectral dependencies on fundamental frequency and sound pressure level only*, *without any reference to a dependency over time*. This can be taken as a starting hypothesis.

The VFM holds a timeless representation. Note that the obvious parallel of a photograph versus a film does not quite hold, as the 'image' is not linked to a modulation, not related to a gesture, nor suggesting one, as a photograph is able to do. Moreover, a VRP picture will not change in the timeframe of a minute. It thus is a semi-disembodied representation that relates primarily to the mechanics of the musical instrument, and only secondarily to the skills of its player.

2.6 Time dependency versus dependency on fundamental frequency and SPL

It is tempting to compare this VF mapping to a lossless transformation procedure such as the Fourier transform. With both procedures, there are different domains with a change in coordinates. In case of the VRP mapping, a dynamic change in voice quality that first proceeds over the time line, gets next rearranged along two new coordinates f_o and SPL that allow a different interpretation of the original information and its inherent dependencies. The resynthesis described in the previous paragraph, demonstrated that this process can be reversed without information loss on the *quality* of the voice. However, all time connected information, like the melody, will be lost. There are some crucial differences with going from time domain to frequency domain or vice versa as with the FFT. With the Fourier transform, the frequency axis links to an unremitting periodicity over the full time stretch of the connected analysis period (the time window). With the VF mapping, the sampling of the time domain information is not concerned with any ordering or dependency over time. What appears to be neighbouring information within the VF can be information that was/is sampled at any moment in the past or the future. This is a non-trivial distinction. Moreover, there is no fixed time frame with the VRP.

The implications are that the amount of change that is observed along the time line is no longer a measure for the amount of change that is observed for the same

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[1] The harmonic envelope concept is described in paper **E**. The resynthesis procedure was demonstrated live, at PEVOC 8 in Marseille (2011), during the workshop: Voice Synthesis from the VRP. A bandlimited oscillator (that allows both FM & AM), drives the model [40],

quality aspect when mapped over the VF. For example, a voice parameter like 'jitter' may show an unsystematic spread when its behaviour over the time line is followed for several seconds. However, this spread may be considerably lower, and may even show a systematic order when mapped over the VRP. Dependencies and variations that appear over time present no clue, and are in no way predictive for the dependencies that are observed as a function of f_0 and SPL. Thus, *VRP mapping may reveal fundamentally new systematic dependencies that will are resolved as systematic effects when observed along the time line*. This is why it is interesting to map the voice field.

2.7 Perception of the vertical scale

If swell-tones are produced at constant pitch during a VRP recording, sound spectra will be stored over a contiguous vertical region in the VF. The spectral evolution as a function of SPL that was sampled during the swell-tone can then be resynthesized. Due to a technical adaptation, a peculiar perceptual phenomenon will appear.

> At a certain point in the technical development of the Voice Profiler system, an automatic gain circuit was implemented. This was done to compensate for large changes in SPL in the output signal that occur during resynthesis. A normalization of the spectral power sum guarantees a constant intensity for the resynthesis, whatever the vertical coordinate position. Although the vertical dimension of the VF is not reproduced in the actual SPL, remarkably we still hear the resynthesized voice getting softer and louder. We still perceive 'volume' changes while the sound intensity remains constant. When the information stored for high SPL is selected for resynthesis, a voice sound with a high effort level is perceived, that resembles the sound of a loud voice. We experience the sound of a very soft voice when the information around the threshold of phonation is resynthesized. When checked with a dB-meter, the SPL will be the same in both cases. This SPL-normalized resynthesis does not sound unnatural; the ear fluently accommodates and thus appears to be quite oblivious to SPL. Listeners have to be told about this mechanism being active and only then become aware of it. The compressing effect is audible, but not as a distortion.

The message inherent to this phenomenon seems to be: *to judge the sound level, the ear is able to rely completely on secondary information, where it could even be so, that the ear does not judge a sound level at all.* (For the moment, there is no demo option to gauge this effect yourself.) The above observation, valid or not, had a great influence on the thinking on the perceptual value of the vertical scale in relation to the VF mapping paradigm. The following list of bullets sketches some considerations and related ideas.

- Consider this apparent contradiction: (A) the SPL is totally irrelevant for the perceptual characterization of voice quality, and (B): the SPL dependency is a central factor in the perceptual characterization of voice quality. Next, rephrase (B) as: rating of the effort level is a central interest in the perceptual characterization of voice quality. Thus, the contradiction is relieved. With the coverage of the full SPL range, a vocal effort continuum is created that perception evaluates in its own way.
- Do we actually have an inbuilt reference to judge the absolute SPL, or do we always rely on the perceived quality of the sound in relation to the always present acoustical context? Note that with any sound we hear, there is always an associated signal-to-noise ratio that could inform us on the absolute level, without the need for an inbuilt reference. Having dealt personally so long with the technical problem of maintaining a fixed absolute SPL when making voice recordings, and considering the extensive hard and software solutions needed to guarantee an absolute SPL in automatic VRP recording systems, my doubt that evolution has felt the same necessity, and resorted to a comparable loop to solve this problem for the ear, only grew stronger with time. Note that we cannot simply assume that the principles: –more hair cells are excited with more power– and –a higher firing rate with more power– will thus mean that we probe sound levels in an absolute sense, as these mechanisms offer a dynamic range of only 30-40 dB [41][42].
- The *perceived* voice range profile or *P*VRP, a concept introduced by Hunter, Švec and Titze [43] will give us a vertical readout that presents a better match to the perceived level. However, the ability to measure something along a vertical scale that does not align with perception, but that does align closely with production, e.g. physiology, is perhaps the strongest selling point of the VRP and the whole mapping paradigm.
- There are some familiar phenomena to mention in this connection. (a) We cannot for ourselves tell if we actually produced the highest SPL for a certain pitch, or if our voice truly gained or lost power; we always need an external sound level meter to confirm it. We have no long-term memory for SPL. (b) Voice experts that do not know the VRP generally have no idea that the SPL is such a precise control for the bounds of a voice register. (c) We are unable to sketch the precise shape of a VRP in

advance. With every VRP recording there will be an element of surprise. (d) When an assessed VRP is questioned, the SPL readout will always be doubted first. The VF-mapping allows a very precise comparison with a previous quality recorded at a precisely matching production settings due to a reliable absolute SPL measurement. Over the years, this message has had disappointingly little impact. People are unable to reliably judge what information an objective SPL measurement could contribute. They have no idea how prescriptive the SPL scale can actually be for the quality of the voice; the SPL remains an artificial dimension.

- The SPL-normalized resynthesis described above demonstrated that all listeners, over their life time, must have gained a sure notion of how the voice spectrum typically changes with increased vocal effort. The way a voice gains power largely characterizes its function and quality. We all know this intuitively, it is a primary interest in our evaluation of a voice. It is a dependency that is not easily quantified. Fortunately, in a VRP recording the related perceptual gradient or perceptual contrast is precisely captured as an objective SPL dependency. The synthesis from the VRP suggests that we are even able perceptually to identify an individual on base of this dynamic gradient (although this remains to be formally shown). That this SPL related gradient is such a fundamental notion is reflected in the choice of spectral metrics that are presented in paper **E**. Each of the metrics focusses on a different aspect of the spectral power development, and each could reveal an increase in vocal effort.
- The SPL scale can be interpreted as a temperature scale that rates the total energy contained in the voice production system. As long as a constant power is radiated from the mouth, and a constant portion of this power is probed with a microphone, the SPL will reflect this temperature, or energy level at which the voice mechanism is operating. This analogy can be taken further. Making a VRP recording, and purposely exploring the vertical axis, can be interpreted as form of ‘voice calorimetry’. Thus, a connection to a thermodynamic model can be made, where concepts from statistical mechanics can be applied in order to understand how regimes with different characteristic, constant exponential factors could emerge, and why the proportional effect of the subglottal pressure could upscale. These principles are briefly introduced in the last chapters.

3 EXPONENTIAL SAMPLING

3.1 Identifying sources of variation: craft or science?

The purpose of compiling a thesis such as this is to demonstrate a degree of scientific competence. However, measuring voice quality is more art than science, where the word 'art' typically means craftsmanship. It is the knowledge of the rationale behind the tools that turns craftsmanship into engineering, or engineering into craftsmanship. As mentioned in the introduction, measuring the voice is modelling the voice, and with that, engineering turns into a natural science.

It might appear that this thesis is primarily concerned with the engineering of recording systems, but the scientific insights still determine the foundation, and provide the understanding behind the model. An unconventional scientific vision defines the measurement paradigm, and it is much more than merely a basis for a test protocol for producing a VRP or SRP.

The measurement paradigm is seen here as a scientific instrument. Still it remains an *art* to close in on the quality, to define the quality metrics, and most essential, to understand and identify the sources of variation in order to *isolate the wanted variation*, and to *manage the unwanted variation*. This last expertise or craft is hard to communicate or to publish on. The expertise is not just a form of 'good laboratory practice' or about the ruling out the sources of variation by clever conditioning external factors. I submit that this art of understanding the origin of the variation is a form of basic science. The answer to any question on the origin of the variation will eventually touch on a fundamental aspect of voice production. If this ground is not touched, if the voice is there for the model, science will miss its promise. This chapter is about the art of aiming for a fundamental quality property of the voice, and it all connects to a method called exponential sampling.

Exponential sampling is a technique that forces you to do the statistics before the actual measurement. Strangely, there is no variance assessment afterwards, but only a conception of variance beforehand.

Note that we are *not* discussing sampling in the time domain. Exponential sampling does not imply a constant exponential increment over successive sampling instants (as might be suggested by the graph in Figure 9, below). The reader is asked not to associate this sampling process with the familiar sequentially-clocked time domain, analog to digital sound signal sampling. The exponential sampling principle is concerned with making observations at any (random) point on any type of scale. Chapter 4 explains why this type of sampling is not just plain scale warping.

3.2 General principle

Exponential sampling and its associated logarithmic scale mapping present a principle for identifying the source of a variation, and for validating that it is the voice that is being measured and not an aspect of the system or of the environment. With the certainty that the voice is the source of a variation, the observed result no longer needs to fit a predefined model, and thus, unpredictable or supposedly unordered information, which so typically characterizes the individual quality of a voice, can be assessed and compared.

With exponential sampling, the representations of proportional dependencies are linearized by putting them on a logarithmic scale. The main interest when recording voice parameters in the VF is to reveal the exponential factors related to these linearized dependencies. This simple idea forms the heart of mapping paradigm. The same proportional binding principle can be used to understand how different macroscale parameters control the voice synthesis, which is subject of Chapter 5.

3.3 Is this a new notion?

In the log/log coordinate system of the VF map, a similar VRP contour shape implies that two voices share the same proportional dependencies (the same exponential factors) in their voice production mechanism. This is not the central message of earlier studies that report on the VRP. Unfortunately, this notion is not explicitly put forward in any of the included papers, either. For instance, the FD-modelling in paper **C** models the variation equally in a 2D vector space, but the point that that the model relies on modelling this co-variation of log/log scales is never made explicit. Nevertheless, this notion was there from the beginning, and it controlled the design of the Voice Profiler system and its software. The notion became the central message when I realized how much the log-scale mapping, that is typical to the VF-mapping paradigm, contrasts with the prevailing approaches in acoustical voice quality measurement. Let us start by reformulating the first sentence of this paragraph: In the log/log/log/etc. coordinate system that is used with the VF mapping paradigm, a similarity in metric distribution shapes implies that voices share the same proportional dependencies (the same exponential factors) in their voice production mechanism.

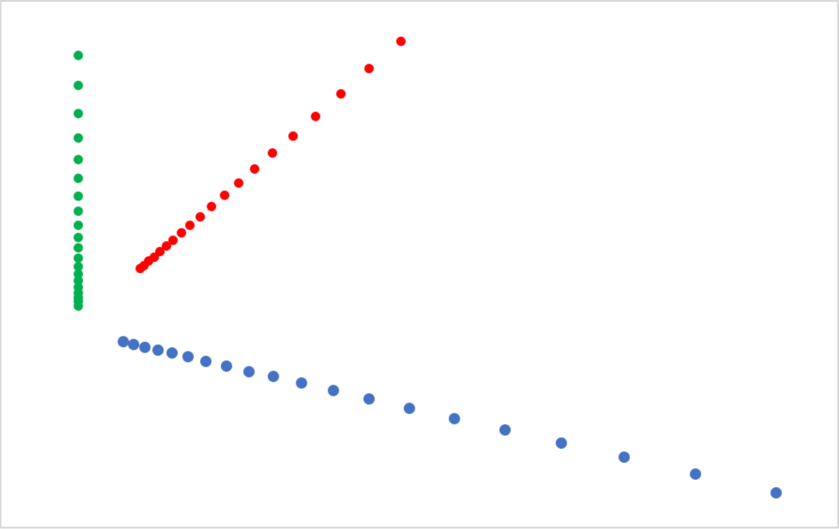

Figure 9. 3D-projection of exponential sampling on three different, orthogonal linear scales.

3.4 Exponential or logarithmic?

Exponential sampling creates a representation on a logarithmic version of the corresponding linear scale. So, if a scale is in linear frequency *f*, the exponential sampling procedure presents a representation on a logarithmic scale that is denoted as: *log(f)*. Although this is generally referred to as a log-scale representation, this should not be called a logarithmic-sampled version of the scale. Why not? Logarithmic sampling would mean that the step size of the sampling interval would progressively decrease with larger scale values. Logarithmic sampling is a procedure that no one devises, for the simple reason that it is not wise to nibble with smaller bites if magnitudes get progressively bigger. When magnitudes get bigger, you want to handle larger chunks, and that is what exponential sampling expedites. So, exponential sampling is the preferred scale warping strategy. The procedure takes a linear scale and turns it to a new ordinate or ordering: the log-scale.

3.5 Recording system perspective

The exponential sampling principle completely defined the software design for (A) the parameter processing, (B) the display and (C) the storage in the voice recording system. The principle largely determines the implementation of the spectrum envelope modelling procedures for which the results are reported on paper **E**. The principle dictates the pitch-period synced sampling model, the scheduling, the data buffering, and it guarantees that the tracking works for all voices over a large dynamic range. The associated exponential variance model forced the development of dedicated nonstandard up-sampling/interpolation procedures. It motivated the development of the watchdog module and assists in the background in controlling the source selection of the dual microphone headset. The focus on modelling exponential dependencies rippled through in the schematics of the dedicated hardware, the choice of electrical components, and even in the layout and discrete changing widths of the wiring on the PCB-layout that are each connected to an on-board created exponential scale in supply voltage levels. All this served to preserve a maximum signal-to-noise (SNR) ratio from the very start and to guarantee the largest possible exponential sampling range for the main ‘temperature gradient’, the sound pressure scale. It is fair to say that exponential sampling became something of an obsession.

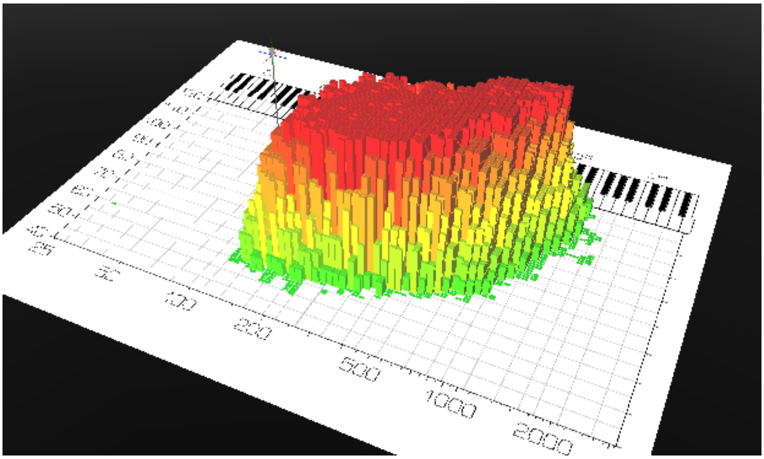

Figure 10. 3D-projection of a voice field with log(f_o) in Hz along the X-axis, SPL in dB along the Y- axis, and with the logarithmically scaled parameter value projected along the Z- axis, and also as a green-yellow-red colour gradient. (cover picture of the Voice Profiler user manual [11]).

3.6 Exponential sampling on all VF scales

With the VF-mapping paradigm, all scales are exponentially sampled by default, and thus they are all log scales (see Figure 10). The X-axis in semitones shows the fundamental frequency f_o linked to a discrete cell index on a log-scale, the Y-axis in dB SPL shows the amplitude (or power) of the sound pressure linked to a discrete cell index on a log-scale, and the Z-axis (or the colour scale) displays the magnitude of a voice metric as a (cell-binned) sample on a log-scale.

Figure 9 showed the exponential sampling on the three linear scales associated with the VF. The dots on the X-axis (blue) depict how the linear fundamental frequency scale is exponentially sampled, the dots on the Y-axis (red) depict how the linear power scale (squared sound pressure) is exponentially sampled, and the dots on Z-axis (green) depict how generically any type of linearly scaled voice metric will be exponentially sampled. Note that on any of the linear scales the zero point will be approached, but never touched (See Figure 9). As a consequence, there is no absolute scale marking shared by the linear scales (which is typical of a so-called affine space [44].

The 3D projection of the 3 axes in Figure 9 tries to mimic the 3D projection of the axes in Figure 10. Hopefully it is clear also that when the same sampling points from Figure 9 are plotted equidistantly, as would done in the cubic VF cell space from Figure 10, all scale marks will reflect a logarithmic scale dependency, where for the moment we ignore differences in scaling factor and range that are associated with each scale.

3.7 Exponential sampling and perception

Exponential sampling of scale information, where steps of equal proportion are represented as equal distances, links to a fundamental property of nature to which perception has obviously adapted itself. Our ability to perceive a difference on a scale, is generally proportional to the scale value itself. This property is seen in many modalities of perception: our sense of roughness of a surface, our sense of touch, critical bandwidth, our sense of pitch, of sound intensity[2]. A logarithmic scale, that reveals an exponential sampling mechanism, is seen with the decibel and the semitone. In his illustrious book: ‘Fractals, Chaos and Power Laws’, Manfred Schroeder [45] calls the logarithmic scale the ultimate self-similar scale and presents many different connections with nature and perception. Exponential sampling is obviously the best choice for perception to deal with nature. If we learn about the world that surrounds us, we combine the observations that our different senses present of the same event. Generally, the

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[2] Prof. Plomp repeatedly poured this inspiring notion over all his students.

reproducible factor that we perceive with our senses scales proportionally with the event. Are we mimicking perception by simultaneously evaluating different proportional dependencies within an ensemble of voice parameters?

3.8 Magnitude range covered in exponential sampling

A close re-inspection of again Figure 9, will make clear that exactly 20 dots are plotted with each of the three scales and in all three cases the sample-spacing increases by a factor of 10 (one decade). In the actual VF-mapping the exponential factors, the *scale parameter*s, that are connected to a sampling grid will be different with each scale. The exponential magnitude ranges covered on the scales differ also.

The largest magnitude is covered with the SPL scale, that in the graph spans a range of 40 to 120 dB SPL (8 decades or a factor 100,000,000 in power). The magnitude range is even larger if the actual sound pressure range which is reserved for the metrics is taken into account. In the VP hardware implementation, there is 6 dB headroom to 126 dB SPL peak level, while the lowest SPL resolved in the narrow band spectrum analysis finds a floor at -30 dB SPL (see paper **E**). This amounts to a remarkable (yet proven) range of 156 dB (15 decades in power, or a factor 63,000,000 in amplitude). Note, that the -30 dB SPL is the power observed within a narrow band of a 2k FFT. Would the 'white' background noise be integrated over the full audio bandwidth, the floor would go up 33 dB, and the total dynamic window would be reduced to 123 dB, which is still a huge amplitude factor.

As for the horizontal scale, the f_o tracking saturates at the high end at around 4 kHz. The tracker does not detect frequency, but excitation events, which could even occur at a rate of only two or three per second. If an f_o of 2 Hz is taken as a low bound (the zero-f_o scale mark will never be reached), the scale parameter for the exponential sampling on the f_o scale is about 3 decades, or 10 octaves (the range f_o < 50 Hz is just not displayed).

It is important to be aware of these very large ranges in order to grasp the idea that VF mapping is about exploring exponential factors, product relationships, reciprocal relationships, shared proportions. So, it is clear that the VF-paradigm is based on log-scale mapping, but why? What is the rationale, why should we use such a mapping system?

4 EXPONENTIAL VARIANCE

4.1 The caveat

Exponential sampling is not just plain scale warping, there is a caveat. If it were plain warping then the previous chapter headings would simply be referring to a ‘multiple log-scale mapping’. The recurrent heading ‘exponential’ is chosen to stress the fact that there is an exponential sampling procedure connected to a scale warping scheme. The caveat is the assumption that the spreading principle connected to the sampling just follows the scale warping from the linear to the log domain or vice versa.

The connected arithmetic process that creates the distribution, does not automatically change identity with a conversion from a linear to a logarithmic scale. An average that is built by summing values on a linear scale, will not automatically become an average product. Such an average product—the N-th root of a signless product of sample values—is called a *geometric* mean [46]. A standard deviation as an average difference from a mean on a linear scale, will not automatically warp with the scale and become a *geometric* standard deviation that describes an average dissimilarity in multiplication factor observed in relation to a geometric mean.

So, the distributing process does not change identity with the scale warping.

To get a geometric mean, or geometric standard deviation on a logarithmic scale, we need to implement the procedure to calculate this, with its particular arithmetic, while still on the linear scale. Moreover, if samples that are already on logarithmic scales are averaged—as could be done by in-cell averaging VF metric distribution patterns—then the nature of the resulting distribution does not reflect an additive or subtractive property on the linear scale.

We came across this issue in paragraph 1.6.1, where it was explained that the jitter, the standard deviation in the f_o that is observed on the linear frequency scale, is by itself a reflection of the individual voice quality. When logarithmically-scaled jitter values are averaged per cell, then these personal *scaling factors for the variance* are averaged. An average log-jitter is conceptually different from an average linear jitter; the first refers to a geometric variance, and the second to a linear variance. The geometric variance for the series of jitter values {1%, 2%, 4%}

and the series {0.5%, 1%, 2%} will be the same, but the linear scale variance will be different.

4.2 Sampling variance in the VF

The general intention with the VF-mapping paradigm is to trace how sampled information will jointly distribute over multiple log-scales. All mutual dependencies that are seen over the multiple log-scales represent a purely magnitude-based (signless) connection. The central aim is to discover shared magnitude scaling factors. A sampled metric is always part of a meta-sample, the result of a *parallel* exponential sampling of magnitudes on different observable scales.

This log-scale world is a realm that is fundamentally different from the linear one. It is a domain with a different algebra that is applicable only by virtue of the fact that the scale is sampled. The log-scale based dependencies are defined only on the sampling points. This discretization is a necessary condition for revealing shared exponential factors over different scales. There is zero information in between all the samples that are mapped over the VF. The log-scales offer no guarantee that there will be a connection between the samples. All metrics are optimized to meet a maximum entropy criterion, implying that all values have the same probability of appearing anywhere on the log-scale. Thus, any continuity suggested in a totally log-scaled world is a conjecture on a multiplicative relationship, without referring to an additive or subtractive affiliation on any of the linear scales.
Note, that the maximum entropy criterion specifies a uniform probability on a *logarithmic* scale and not on a *linear* scale. According to this notion, pink noise is the true statistical white noise, because it has a flat distribution per interval on both the log(t) and the log(f) scale. Pink noise is the ideal test signal to reveal if the maximum entropy remains conserved under frequency transformation [58].

In statistics, it is customary to rate the error in a measurement by evaluating the standard deviation that is observed for an expected value. This is typically a post hoc process where N samples are evaluated; where the N matters. However, for the f_o and sound pressure scale in the VF mapping paradigm, the standard deviations (characterized by the jitter and the shimmer) are modelled as continuous time signals. These aperiodicity signals are treated as information carriers that can be instantly sampled together with the mean f_o and the mean pressure signals at the meta-sampling period-clock instant. The jitter and shimmer signals are modelled (interpolated) such that they measure a geometric deviation from the geometric mean f_o and the geometric mean pressure.

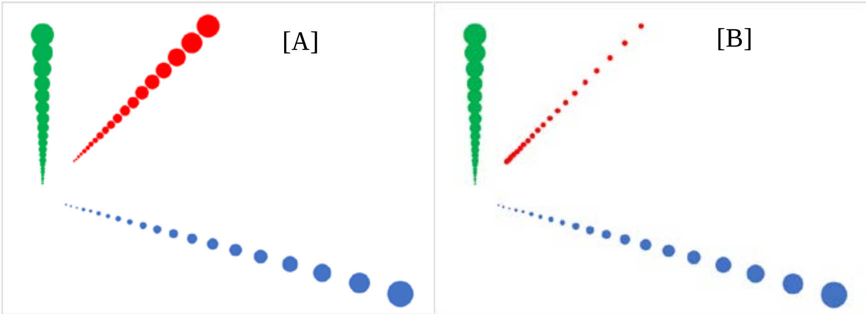

Figure 11. 3D-projection of an exponential sampling process. [A] The point size reflects a proportional increase in the standard deviation (a constant coefficient of variation) on all three linear axes (blue: linear frequency, red: linear sound pressure, green: any other linear metric) [B] the constant point size along the red (pressure) axis represents a constant standard deviation, induced for instance by a constant quantization error or a constant external noise source that adds an error to the sampled linear sound pressure.

Figure 11 is meant to pose a little conundrum, to make several points clear. The positions of the blue and red dots on the linear scales mark a mean f_o and a mean sound pressure that, when exponentially sampled, determine the log(f_o) and SPL cell coordinates in the log-scaled VFM. The size of the red or blue dots symbolises a standard deviation on a linear scale. At the sampling instant, this spreading factor becomes detached from the X and Y scale and will be ported to another scale, a quality metric scale (green). The uncertainty in the X and Y directions (the jitter and the shimmer) will no longer show as a dislocating effect on the original scale. This is because the *mean* f_o and the *mean* sound pressure are exponentially sampled. This X/Y scale spreading information can again be mapped as an independent jitter or shimmer quality metric *distribution shape* on the vertical (green) scale.

An external acoustical noise source may define a floor for the lowest mean sound pressure. With a high average sound pressure coming from the voice, the effect on the overall (mean) SPL will be negligible. However, a shimmer metric that measures the random deviation in peak-to-peak pressure over consecutive periods, is able to sense this small noise contribution. The standard deviation related to this perturbing effect is pictured in Figure 11-B by a constant-sized red-coloured dot on the linear sound pressure scale [but not on the log scale]. This

constant linear standard deviation that was first connected to the red scale, next repetitively maps to the same green dot on the vertical quality metric scale that reports a constant standard deviation.
It is typical of the voice that the standard deviation in the peak-to-peak amplitude increases proportionally with the amplitude of the peaks; the shimmer in percent remains constant; a constant coefficient of variation or constant relative standard deviation. If next this shimmer in percent is mapped as a linear percentage along the vertical scale, this constant percentage would be marked by the same-size dot on the vertical scale. A proportional change in shimmer percentage will appear as a constant linear distance for the log-scaled metric. But shimmer whose origin is not vocal will be represented differently.
Note that all metrics may assume any value anywhere in the VF, yet the associated spread on the log-scales will not reflect an ordinary sampling error for that metric. An unsystematic deviation that seems to contradict an apparent log-scale dependency can be interpreted as a sign that entropy can be manifest anywhere on the log-scale. If the environment does not predict the metric value, the distributing process may corroborate its external (vocal) origin. Remember, we must be wary of smooth distribution shapes.

4.3 The log-normal distribution

The localizing principle explained in chapter 1 is based on the practical observation that the numerical value of any recorded voice parameter is remarkably well determined by the cell coordinates in the VF. Given one observation in a particular cell, the expectation is high that subsequent observations in that cell will give similar results. Although it is clearly stated in chapter 1 that an in-cell variance calculation is not supported, it should still be possible to calculate such a value, and to link a standard deviation (SD) value to the observed mean. What would happen if we were to be less dogmatic?

Note that all sampled information would be averaged on logarithmic scales, and not on the original linear scales. A logical thought is that the distributing process on the logarithmic scale follows the central limit theorem (CLT). One may thus expect that the CLT applies for all the logarithmic metric scales that are used in the VF-mapping. Samples would be expected to distribute normally on each of the log-scales.

The associated distribution model is called a *log-normal* or Galton-distribution. With this distribution model, all samples are considered to be observations of a single random variable. With the averaging performed on logarithmic scales, the resulting mean thus represents an average multiplication factor (or average exponent). The average product to which the mean on the logarithmic scale refers is called the *geometric mean*. This average exponent can alternatively be interpreted as an average logarithmic probability. Probabilities combine in a multiplicative manner, and on a logarithmic scale this operation becomes an additive operation.

$$PDF(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(ln(x)-\mu)^2}{2\sigma^2}} \qquad (4.1)$$

Formula *(4.1)* describes the probability density function (*PDF*) for the log-normal distribution as observed on the original linear (x) scale [47]. The μ and σ describe the mean and standard deviation that would be observed on the logarithmic scale. Note that seen on this linear scale the distribution is not normally distributed and thus μ does not coincide with the peak of the PDF. Also, σ does not effectively describe the width, as with larger σ the log-normal distribution skews to the left when observed on the linear axis (see Figure 12).

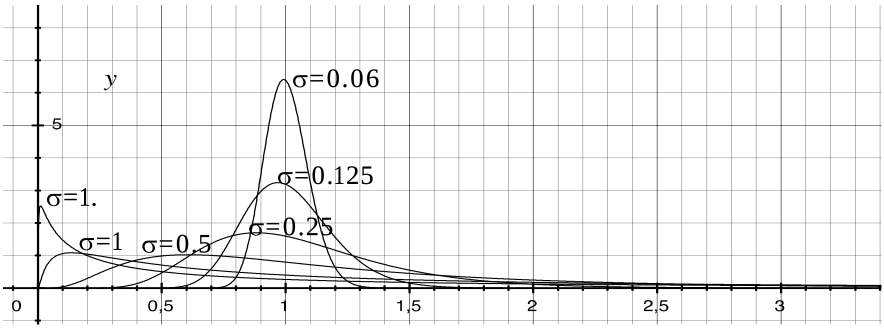

Figure 12. Probability distribution functions (PDFs) following formula (4.1 with μ=0 and σ= {1.5, 1, 0.5, 0.25, 0.125, 0.0625}. For small σ values, the Gaussian shape will be approached on the linear scale too.

By rewriting Formula *(4.1)* to Formula *(4.2)* the reciprocal dependency that scales the Gaussian is brought inside the exponential function.

$$PDF(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[-\frac{(ln(x)-\mu)^2}{2\sigma^2} - ln(x)\right]} \qquad (4.2)$$

As the exponential sampling transforms a linear-scale into a log-scale dependency, the derivation of the corresponding log-scale based $\mathrm{PDF}(x_l =$

ln(x)) for the log-normal distribution is now straightforward (see Formula *(4.3)* and Figure 13).

$$PDF(x_l = ln(x)) = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[-\frac{(x_l-\mu)^2}{2\sigma^2} - x_l\right]} \qquad (4.3)$$

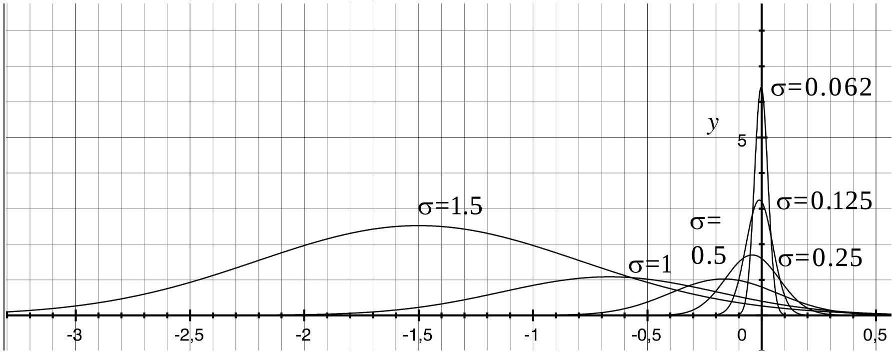

Figure 13. Reshaping of the (PDFs) from Figure 12, when expressed on a logarithmic-axis: x_l, as described in formula (4.3) with again σ= {1.5, 1, 0.5, 0.25, 0.125, 0.0625} and μ=0. Note that σ controls both the width and centre (mode) of the normal distribution that appears on this log-scale.

The above graphs show that the spread observed on the linear scale, (characterized by the arithmetic standard deviation s), and the spread observed on the logarithmic scale, (characterized by the logarithm of the geometric standard deviation σ), are indirectly related. The two spreads are bound by a variance relationship described in Formula *(4.4)* where the variable m denotes the linear-scale based (arithmetic) mean.

$$\sigma^2 = ln\left(1 + \frac{s^2}{m^2}\right) \qquad (4.4)$$

Both σ and s characterize a width of a distribution on their connected scales, but the width on the logarithmic scale links to a normalized width on the linear-scale. This normalized width, that appears as a squared factor at the right-hand side in formula *(4.4)*, is called the 'relative standard deviation' (RSD) or 'coefficient of variation' (CoV). It is the ratio of the arithmetic standard deviation s and the magnitude of the arithmetic mean m. This means that the width of the log-normal distribution reflects the coefficient of variation. Formula (4.5) describes how the logarithm of the geometrical mean μ relates to both the arithmetic mean m and σ

(seen on the linear scale). It explains the offset change for the maximum (mode) observed in Figure 13.

$$\mu = ln(m) - 0.5\,\sigma^2 \qquad (4.5)$$

4.4 Realism of the log-normal distribution

The Gaussian shape on the logarithmic axis that characterizes the distribution as log-normal is always the result of an averaging procedure subsequent to an earlier exponential sampling process. It is not possible to directly sample from a logarithmic scale, as such a continuum is non-existent in the physical world. So, if a log-normal distribution occurs, then nature is not providing a 'hardware' implementation of a log-scale to sample the distribution from.

A Gaussian distribution ultimately emerges when the CLT acts on a linear physical observable scale. The Gaussian shape reflects the circumstance that each new observed value has a random term that is not due to the observed entity, and thus represents a deflection (with an energy) in a dimension that is orthogonal to that of the scale. The total amount of orthogonal deviations (or unexplained energy) is measured simply by summing squared deviations from the mean, and this yields the variance. In this sense, the Gaussian shape is a sign that there are one or more sources of variation that are unrelated to the measured entity.
However, a squared distance criterion is not native to a log-scale. Log-scales do not support a reference to a linear scale zero point for which an orthogonal deflection can be evenly scaled. There is no shared zero energy reference or linear scale mark that can be generally referred to. How can the CLT show its effect on a logarithmic scale, when there is no statistical principle on the scale to support it?

This makes the log-normal distribution a rather treacherous template. However, we can use a model that complies to a linear-scale standard deviation concept, by modelling a generalized geometric average deviation, and leave the possible emergence of a Gaussian distribution shape on a log-scale to the people that still need it as a template. Since a fixed square root operator connects the standard deviation to the variance, the two concepts unite on a log-scale. The problem that nature did not provide a hardware implementation for us to directly access the geometrically scaled domain can thus be solved.

4.5 Modelling modulation

The voice signal is typically a modulated signal, where a temporal modulation on different scales (frequency, amplitude, frequency characteristic) carries the

actual information [48][49][50][51][52]. The fundamental frequency and amplitude are never completely stationary. We discern fast modulations like jitter and shimmer, slower modulations like wow and flutter, and periodic modulations like pitch vibrato and amplitude vibrato.
To analyse the individual contributions of these components, there is the following choice:

(A) model all modulations as being linearly additive/subtractive effects in relation to a linear scale mean, or

(B) model all modulations as being proportional to a geometric mean, and represent the modulation as a physical process where a division by a certain factor is averaged out by a multiplication with the same factor.

Strategy (B) conceives frequency and amplitude modulations as proportional factors that linearize on log(frequency), log(amplitude), log(f_0) scales. If these modulation components need to be segregated, the filtering, too, needs to be done on proportional (exponential) scales. Furthermore, this implies that frequency (or phase increment), too, should be modelled as an exponential dependency, which leads to an exponent in the imaginary part of a complex exponential. Note that the sign of the frequency thus becomes meaningful, as there no longer is mirror symmetry around zero frequency. Poles and zeroes no longer come in pairs, on an exponentially sampled frequency axis. The modelling of the time and frequency domains transforms completely with these conceptual changes [58].

4.6 Statistical analysis by synthesis

The log-scale based averaging is associated with the proposition that our samples exclusively include magnitude spreading effects that result from additive or subtractive *exponents*, while the magnitude spreading effects of additive or subtractive *linear terms* no longer play a role.
An average magnitude is reflected in the energy, or average power associated with a system. If the logarithm of this energy, or average power is not varying, its value could be interpreted as an estimate of the μ, the logarithm of the geometric mean. As argued in paragraph 4.4, there is no physical observable scale from which to sample a log-scale based geometric standard deviation σ. However, a σ-like measure can be modelled by generating a second order exponential (magnitude) based distance measure on a connected up-sampled linear time scale. The synthesis of such an exponential domain as a time signal (a microscale) is a key principle in the implementation of the VF-mapping paradigm. For this reason, complex exponential (analytical) signal representations are used in the statistical processing of all voice metrics in the Voice Profiler system. In principle, the FD-averaging (paper **C**) follows the same conceptual thinking, but not on a time scale.

4.7 Why not show the in-cell variance if the formula is correct?

The σ and μ in the earlier formulae are mainly of analytical importance, as they will *never be directly* observable as the result of a physical averaging process that occurs on a logarithmic axis. The same applies for the arithmetic mean m and arithmetic standard deviation *s* in the same formulae. If m and *s* were calculated in practice by converting the scale for each sample from logarithmic to linear, the principle on which this formula is built would be violated, as m and *s* still refer to an exponent averaging process based on pure magnitudes (and not to an additive process on a linear scale). Warping of the distribution function from the logarithmic to the linear domain will not change the underlying distributing process with which formulae *(4.4)* and *(4.5)* are associated. In a practical implementation, resampling of a distribution will always lead to a loss in information content [53]. Thus, a PDF that is described on a continuous scale represents an abstraction that, without a connected sampling model, can be highly deceptive.

This all sounds like a detail, an exercise in wording, a problem that is effectively covered by stringently following the mathematical rules. However, to me it was, and still is, not a *minor* but a *major* detail, where the mathematics could be formally correct when they remain detached from an actual sampling procedure.

> As a voice recording system developer who is obsessed with exponential sampling, I could not stop wondering about the following question: How can we tell what the shape of a distribution will be on a new scale, if the algorithmic nature, the spreading principle, on the old scale, is not predictive for a distance on the new scale? Is this a problem only because just a single sample is taken into account, rather than a group of samples that may be grouped under a distribution shape? I found such an explanation unsatisfactory.
>
> The same group of random samples that all obey to an *independent* identically distributing (IID) process on a linear scale, can be part of a *dependent* (non-random) identically distributing process seen on a logarithmic scale. In the VF mapping paradigm, we may find a proportional dependency that appears only on log-scales that is not directly linked to any dependency on a linear scale. A multiplicative connection is conceptually different from an additive connection. In the same manner, a group of samples that belong to an independent

identically distributing (IID) process on a logarithmic scale, can be part of a dependency, a dependent identically distributing process, seen on a linear scale. The IID property that conceptually defines a PDF is not automatically preserved with a scale transformation from linear to logarithmic. Thus, a PDF as such, cannot be warped from a logarithmic scale to a linear scale when the same statistical concept cannot be fully supported on both scales. Consider a mathematical formula that contains a PDF that describes a statistical process, then there must be a connected sampling process. Disconnecting such a formulated abstraction from a sampling process, by assuming a continuous scale for probabilities, puts you into a conceptual danger zone. The correction factor for the predictable density change on the continuous scale, does not solve the issue for the independence. In my perception, the problem associates to a basic concept from quantum mechanics, where each sample remains paired with its distribution, that defines its chance to be observed in a certain state. The distributing process is characteristic to the sample, and the sample does not have an identity without an evaluation of the distributing process. I here take the precaution of stating that my obsession with this problem of warping PDFs might be relaxed if I were able to change my point of view, and had a better mathematical understanding to see potential errors in my thinking.

4.8 Comparing probabilities

The VRP and SRP are not stored as contour curves; that abstraction is reached by post-processing the phonation density distribution (see paper **F**). The VRP and the SRP can both be seen as a two-dimensional histogram, a 2D statistical distribution. By normalizing for N, the bin count for all densities in the VF cells, such a distribution can be transformed to a 2D probability distribution, that specifies the probability of arriving at a given cell position in the VF. With this probabilistic interpretation, it becomes possible to connect conceptually to the ideas from information theory [55], and to calculate for instance an entropy for a SRP or VRP distribution. Moreover, the VRP and SRP are actually 2D *conditional* probability distributions, and thus the concept of mutual information [55][53][54] can be applied to test for independency of scale and to derive an objective measure that assesses a similarity in distribution shapes. For such a similarity measure, there is no need to comply to a specific distribution shape model. Moreover, a *running* entropy measure can be implemented to specify when new speech information is actually redundant, when it no longer changes the shape of the SRP distribution. Perhaps someone else will pick up on these ideas. However, the VRP and SRP are not the main topic of this framework text, and I first intended not to introduce these concepts here, nor to dive into entropy.

However, the comparison of abstract voice quality metric distributions is a very relevant issue in this thesis. Still, there is one main difficulty: the voice quality metrics in the VF do not rely on N, the density count. They are designed to be instantaneously at their target value. Metrics do not disperse with a certain probability, nor is there a predefined probability for a metric associated with a VF cell. The recording system is optimized to not show any preference or bias to any preconceived mode; but rather to comply to a maximum entropy model. However, without a probability, it is not possible to apply the concept of mutual information, or to test for redundancy. It seems that the entropy concept is very useful here.

4.9 The Connection

When Shannon singlehandedly[3] founded information theory [55], he defined the entropy as a sum of information units, expressed in *bits* (using log2), in *nats* (using the ln), or *hartley's* (using briggs log10 base) [56]. Each unit of information $I(p_i)$ is a function of a discrete probability p_i and the total information is described by his famous formula for the entropy $H(X)$:

$$H(X) = \sum_i -p_i \, log_2(p_i) \quad (4.5)$$

Shannon also derived an entropy description based on a continuous probability scale where he changed the sum to an integral and replaced the PMF (Probability Mass Function) by a PDF. Shannon's ideas were improved upon by E.T. Jaynes [60][61], who demonstrated the explicit need for a partition function to convert Shannon's entropy expressed as sum, to match to an entropy expressed as an integral. Jaynes explicated that the use of PDF without a sampling function may lead to an incorrect result!

> This confirmed for me that my ongoing fixation with the PDF warping problem was not that strange an obsession. Reading more about this, I found out how Jaynes formalized the fundamental principles from statistical mechanics, and how he connected the physical definition of entropy and Bolzmann's principles with Shannon's entropy. Jaynes argued that it is possible to apply the

[3] I needed to copy this appropriate qualification when I read it on the Wikipedia page.

same theoretical principles detached from a physical interpretation, and to use these principles for stand-alone statistical inference. He mentions that the same concept can be applied as an abstract statistical modelling technique and that it formalizes the aim to harvest the most information (the maximum entropy) from observations of an abstract system. The explanations of E.T. Jaynes paved the way for an overdue understanding of what I was actually implementing as a VRP recording system developer. Jaynes made me understand the connection between exponential sampling and the partition function, and I realized that the VF mapping paradigm had found a theoretical homeland. Entropy turned out to be more than a handy concept to evaluate SRP and VRP density distributions. The same concept presents also a statistical mechanics interpretation that makes it possible, under the VF mapping paradigm, to make various predictions about the system behaviour using the microscale and macroscale concepts. Being familiar with implementing physical models, where ODE's are numerically integrated using the Riemann integral, I started to realize that this type of integration had a counterpart in the Lebesgue-Stieltjes integral that is concerned with integrating on-scale probabilities, and I began to see that this type of integration has a closer link to the way perception integrates information. Our brain's task is to make sense of the information that comes in via our senses. Perception can only make sense from what is happening in the outer world, by integrating the combined information it receives from each independent perceptual source. Thereby, the information that varies with the same constant proportional factor (on the log-scale) is most logically connected. The insight that a shared constant derivative on the log-scale, was the factor that connects the information was the main message in the paper that I wrote in 2013 with Jordy van Velthoven for the SMAC/MSC conference, which already conveys a salient part of the message presented here [58].

The exposure to the thinking of E.T. Jaynes created a logical connection. Being trained myself for many years in thermodynamics for my degree in chemistry, combined with Jaynes lecture notes [62], and the excellent Wikipedia tutorials on statistical mechanics [59], and added to that my own degree work on ^{31}P nuclear magnetic resonance spectroscopy (which made me study quantum mechanical concepts), this all helped me to conceive the "balloon in the box" example that you can read about in the next Chapter, and which presents the combined insight.

5 EXPONENTIAL FACTORS IN VOICE CONTROL

In this chapter, the idea is explored that constant proportional dependencies are the shared factors in the parametric control of the voice. The shared exponential relationships that are observed with the VF-mapping paradigm would not be so systematic if they were not due to a generating principle that follows the same paradigm. The same proportional dependency model (a framework of shared exponent factors) provides a possible answer to the complex of questions formulated in the introduction, summarized as: What is the basis for the specificity and reproducibility that can be reached in mapping individual voice quality? If scales are so fixed why is there so little uniformity and alignment on these fixed scales? Why are the distribution shapes only superficially comparable?

5.1 The balloon in the box

The rationale behind the mapping on multiple logarithmic scales is best explained starting from an example from thermodynamics.

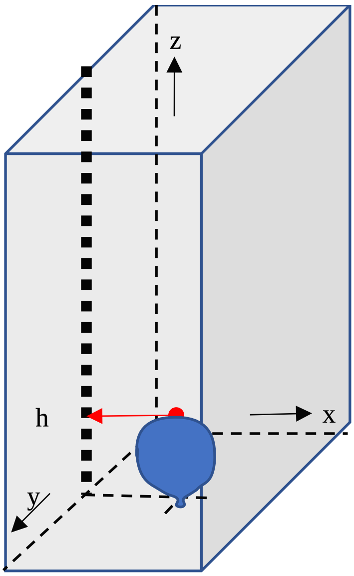

Figure 14. Balloon in a rectangular box as an example of a thermodynamic system, the red arrow marks the height of the balloon from the bottom of the box

Figure 14 shows a balloon resting on the bottom of a rectangular box. The gas in this balloon does not exchange any heat or mechanical energy with the environment. Thus, the balloon can be considered a closed system for which the relationship between the pressure P, the volume V and the temperature T of the gas inside the balloon are described by the universal (ideal) gas law, see Formula (5.1) [63].

$$P \cdot V = n \cdot R \cdot T \qquad \textit{Formula (5.1)}$$

The gas constant R does not change; via Avogadro's number it refers to Boltzmann's constant k_B. The number of particles n, measured in moles, is constant in this example as there is no gas escaping from the balloon. The above formula developed historically as an empirical law. A theoretical physical foundation for it arrived at the end of the 19th century, when Boltzmann connected this initially empirical law to universal statistical mechanical principles. *Formula (5.1)* can be rewritten as a logarithmic dependency where in a somewhat unconventional way, the dB-scale is chosen as the common base, see *Formula (5.2)*.

$$10log(P) + 10log(V) = 10log(n) + 10log(R) + 10log(T) \; \textit{Formula (5.2)}$$

In the thought experiment that follows next, we imagine that we are able to measure three parameters of the system: the height H (in the z-direction) at the top of the balloon, the pressure P and temperature T within the balloon. At the start of our experiment, marked as point S1 in Figure 15, the temperature in the balloon is lowest, and the volume occupied by the balloon is still small compared to the volume of the box, so except for the supporting bottom, the balloon is not touching any wall or the top of its enclosure.

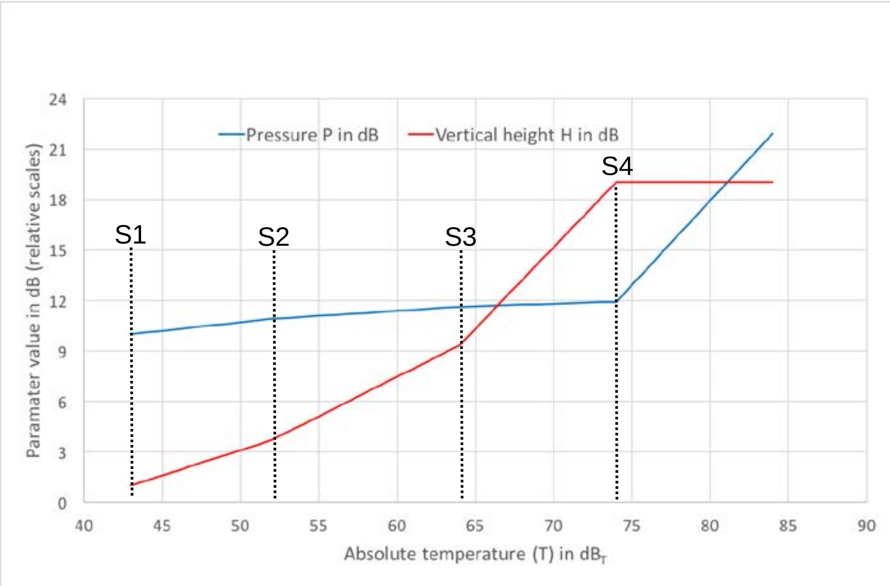

Figure 15. Vertical height in dB, and balloon pressure P in dB as a function of absolute temperature in dB_T (this unit is explained in the text). The breakpoints S mark changes in regime. Note that the pressure and balloon-height curves do not share a common reference or zero-dB point and each have an arbitrary vertical offset.

Next in this thought experiment, energy is fed into the system by gradually heating the gas in the balloon. This process is stopped at the moment that the (absolute) temperature T is raised by a factor of 1.259 (≈1 dB). According to either formulae *(5.1)* or *(5.2)*, this proportional factor will always be equated by a proportional change in pressure and a proportional change in volume (with n and R constant). The proportional factors for P and V will depend on the system settings. The membrane of the balloon acts like a weak spring, and thus the pressure in the balloon can be assumed to increase with only a small proportional factor 1.023 (ca. 0.1 dB) with every 1 dB increase in temperature (next abbreviated as dB_T). This implies that the volume will increase much faster, as it has to make up for the remaining 0.9 dB. As long as the balloon does not touch the walls, the expansion in the x, y and z directions will be proportional. The cubic factor for the volume $V=W.D.H$ (width × depth × height) will be equally spread, and thus we can predict that the height H will take care of a 1/3 of the 0.9 dB = 0.3 dB. When the temperature is raised and no wall is touched, the proportional increase in pressure (0.1 dB) and the height (0.3 dB) in relation to a proportional (1 dB_T) increase in temperature will remain constant.

At some temperature during the heating, marked as S2 in Figure 15, the balloon first touches a wall, and a new regime is entered. As the width of the rectangular box is chosen to have the smallest vertex in this thought experiment, the balloon will deform and flatten perpendicular to the x-direction. The only way the balloon may still expand with an increase in temperature, is in the y-, and z-direction. As the walls that are touched are hard, the area of the membrane that needs to compensate for the increase in pressure, needs to increase faster with increasing volume. Thus, the surface force of the membrane will decrease with a larger proportion and the membrane feels less stiff. The pressure increase in this mind experiment, is now assigned a proportional factor of only 1.014 (ca. 0.06 dB) for a 1 dB_T increase in temperature. This implies that the volume change has to make up for the remaining 0.94 dB. With the width frozen, the volume change is now proportionally spread over the two remaining dimensions, and thus we can predict that the height *H* will take care of a 1/2 of the 0.94 dB = 0.47 dB. When the temperature is raised and neither the ceiling nor the other sidewalls are touched, the proportional increase in pressure (0.06 dB) and the height (0.47 dB) in relation to a proportional (1 dB_T) increase in temperature will remain constant for the regime between S2 and S3.

At some point of heating, marked as S3 in Figure 15, the balloon will touch the other sidewall of the rectangular box, and again a new regime is entered. With an increase in temperature, the balloon is now able to expand only in the z-direction. The area of the membrane that needs to compensate for a pressure increase will even increase with a larger factor. Thus, the surface force of the membrane will decrease with an even larger proportion. The total surface feels again less stiff. The proportional pressure increase in this mind experiment, is now set to a factor of 1.007 (ca. 0.03 dB) with every 1 dB_T increase in temperature. This implies that the volume change has to make up for the remaining 0.97 dB. With the width and depth frozen, the volume change is now compensated by the only dimension remaining. Thus, we can predict that the height *H* will take care of the full 0.97 dB. When the temperature is raised and the ceiling of the enclosure is not touched, the proportional increase in pressure (0.03 dB) and the height (0.97 dB) in relation to a (1 dB_T) increase in temperature will again remain constant in this third regime.

At some point, marked as S4 in Figure 15, the balloon totally fills the box and a final regime will show where the pressure in dB increases in linear proportion with the temperature in dB_T. This is because the volume is fixed and will show no proportional change. At least theoretically there is no endpoint on the scale.

What can be learned from this thought experiment, and what is the connection to the VF mapping paradigm?

- One basic formula where macroscale system parameters are bound together in a product dependency describes the overall behaviour. The geometry of the box and mechanical factors like the stiffness of the balloon membrane, determine how this equation expresses itself dynamically on measurable scales.
- The temperature scale acts as the main reference and is also a measure for the overall energy content (and the entropy) of the closed thermodynamic system. In the VF-mapping paradigm, the SPL assumes this role and acts as the main reference while it is also a measure for the total energy contained in the system. Thereby it is assumed that a constant proportion of this energy is radiated from the system, and from this radiated energy a constant proportion is picked up by a microphone.
- Note that in the regime between S3 and S4, the pressure and height factors add up to 1 dB and take care of the full 1 dB_T change. This is not a proof of ‘closure’, of knowing all the factors. There could always be other factors involved in the total equation because we observe metrics that could be indirectly coupled. Some metrics could even show an inverse proportional dependency on temperature and show a negative sloping in dB/dB_T. Imagine what would happen if the enclosing box was conical. Thus, a proportional increase in volume would lead to an even larger proportional increase in height, so a slope in dB/dB_T larger than one.

  Accordingly, with the VF-mapping paradigm, multiple metric scales are used to observe the system. Each metric is expected to show its own characteristic sloping in dB/dB_{SPL} and there could also be different regimes for different parts of the total SPL range. If the voice follows comparable principles, then the characteristic sloping factors would remain constant within a regime. A separation into three different linear sloping regimes, each associated with a different section of the SPL range, was observed in reality with the SB metric (see figure 9 in paper **E**). Note that the underlying explanations for the mechanisms of the regimes will be very different. We can only guess where they come from. However, the different slopes in dB/dB_{SPL} that are observed in reality indicate that the voice production mechanism in its behaviour obeys an equation where the observed macroscale parameters are caught in multiplicative dependencies. Following the parallel of the virtual balloon experiment, many voice metric scales may reflect different, but still proportionally related dynamical aspects of the system. The metrics do not characterize all the change in acoustic energy, the image is never guaranteed to be comprehensive, as there can always be an unobserved scale.

- It would be an idea to plot the dB_T axis vertically and to introduce a second dependent scale as a horizontal axis and to choose for this scale for instance the box width. The ticks for this horizontal scale could show the proportional change in box width. Note that any log-scale could be used, for instance a scale unit that we could name dB_W, but also a log base 2 (an octave or semitone scale) could be a sensible option. It is obvious that such a choice of coordinate system would imitate the choice of the SPL/log(f_o) scales used in the VF-mapping. The proportional dependencies of balloon pressure and height on temperature could thus be mapped for different exponentially sampled box-widths. With an all log-scale based framework (we could call this the balloon field or BF), the gradients and bending points for the regimes will linearize in two dimensions. Note that the choice of an 'independent' scale will always be arbitrary. In principle, any pair of log-scales may report on a linear dependency that are seen between all metric log-scales within a regime.
- With a different dimensioning of the enclosure, the bending points between the regimes will shift over the temperature scale (expressed in dB_T). With a different membrane stiffness, the characteristic sloping in dB/dB_T that is associated with the regimes will change too. The sectioning of the dB_T range and the sloping between the break points, will only be globally comparable between different sized systems. The pattern of linearized dependencies 'float' on the log-scales, there is no hard dB_T reference point where the dynamic patterns for the different systems will repeat exactly the same dependency. Still, the dynamic patterns all reflect a dependency that is defined by the same underlying formula.

This line of thought can directly be applied to the VF-mapping paradigm. With a different geometry of the voice box and a different physiological dimensioning, there is no hard reference point where all dynamic patterns will show exactly the same dependencies. If the same mechanism applies as with the balloon experiment, this would explain why the voice metric patterns for the individual VRPs are only superficially comparable, while all voices still refer to the same underlying formula or principle. Figure 9 in paper **E** that pictures the dependency of the SB metric on SPL at one specific f_o, shows that most individuals show a sectioning in three regimes, where the position of the bending points and the sloping within the sections are different between individuals.

- The balloon experiment can be repeated by sampling for the absolute temperature in any random order. If the geometry and physical parameters remain the same, then exactly the same pressure and height sample values will appear on the (exponentially sampled) scales as they all link to a reference energy level for the system measured in $dB_{T.}$ In the VF-mapping paradigm, this is the localizing principle, where the absolute SPL scale guarantees the same reproducibility for all its connected exponentially sampled metrics. The reproducibility is guaranteed for any voice and any metric, but as noted before, it does not guarantee a mutual comparability between voices.

5.2 Statistical mechanics

In the previous example, the thermodynamic behaviour of the balloon was completely determined by the gas law, but does that also imply that the voice production system is compliant to this law? Although we are also dealing with breathed air as a gas, where inevitably pressure, volume and temperature remain proportionally related, it also sounds logical that a very different fundamental dependency could be defining the mechanics of the voice production system. Although the 'system equation' may be different, an integrated effect that is seen at the macroscale will still result in a description where parameters are bound together in an equation that involves products (and no sums).

The mapping in a totally log-scale based environment is part of a more general statistical analysis method [59][62] and refers to a more general integration principle. The theory of statistical mechanics, for which a qualitative introduction will next be given, explains how this principle functions.

5.3 Microscale versus macroscale

In statistical mechanics, a system is characterized on two different scale levels: microscale and macroscale. At the microscale level, the mechanical properties of the individual particles are described in equations that characterize the interactive forces experienced by the individual particles in terms of their displacement, velocity, mass, etc. With many particles seen over larger volumes, the magnitude scaling factors for each of the microscale force factors will still be discerned as magnitude scaling factors of an associated statistical distribution of which at the macroscale level only the mean effect is felt. The mean effect felt at the macroscale is a linear combination (an ensemble effect) of microscale scaling factors. The net effect seen for a single microscale physical parameter may average out statistically, but the combined effect (the energy involved) does not

average out and controls the magnitude of the mean effect that is observed as global parameter at the macroscale level. For instance, the average magnitude of the velocity seen for the individual molecules of a gas at the microscale level is reflected as an integrated total kinetic energy factor, for which the temperature parameter that is observed at the macro scale is a measure.
So, the combination of magnitude scaling factors within the ensemble survives the averaging process. However, within this ensemble, the individual scaling factors are no longer discernible as a separate effect. With each upscaling in magnitude of the system, the net effects of the individual scalers will increase with the same factor. Thus, the linear combination of magnitude scaling factors, scales up proportionally. The linear combination survives as an exponentialized linear combination when observed at any given macro scale. So, any mutual relationships or dependency that originates from the microscale ends up encapsulated within the brackets of an over-all exponent at the macroscale.

> The mathematical model that connects the microscale with the macroscale involves a statistical description that uses probabilities to characterize the microscale dependencies. These probabilities are integrated/summed along each of the scales, using an exponential sampling at different magnitude factors. In this formalized sampling, the encapsulating exponential function is called the 'partition function', where it is a Lebesgue type integral that integrates the micro- to the macroscale.

In summary: the force scaling factors that originate from the mechanical model at the microscale end up as a linear combination of factors that as a whole are encapsulated in an exponent on the macroscale. Each macroscale parameter that is observable will reflect its own linear scaled combination of the influences of the mechanical parameters from the microscale. Thus, macroscale parameters appear to be bound mutually in a multiplicative manner, as they are all connected via shared exponential dependency (that reflects the integrated result).
An exponential link will be a proof that all macroscale observable parameters reflect on the same mechanical model as their source. Conversely, the macroscale observable parameters will always connect in this manner when they reflect on the same integrated mechanical system. The repetition of a multiplicative factor on a microscale could stand on its own, but when there are different multiplicative factors repeated in an integrated microscale environment, the connected macroscale parameters will be manifest in a shared exponential dependency. This general principle does not depend on the form of the mechanical system equation seen at the microscale.

5.4 A metric is also a measure of something

The metrics that are mapped in the VF are macroscale parameters that each measures the voice in its own sense. The term 'measure' as an arbitrary scale has a mathematical interpretation too. Typical of a mathematical 'measure' (a Lebesgue measure) is that there is no conception of an associated power or physical dimensionality. Normally, a connected independent scale would be used to characterize the power dependencies where the measure is the dependent scale. However, measures are egocentric and know only their own scale marks and use these as a base to reflect on what is observed along other scales.

We came across an example of a measure when the balloon height was mapped in the thought experiment. The balloon height measure still measures a physical length in meter, but the scale could also reflect on an area in squared meters or even a volumetric aspect (cubic). This was because it was actually reflecting on a dependency linked to a volume. When the balloon started to inflate with increasing temperature, first the width and later the depth dimension were constrained, and the connected power increment factor in dB/dB_T increased stepwise. With a conical box, the associated power increment factor for the height could become more than one, as in order to match the increase in volume, the height needs to increase even faster.

Let us now return to the voice metrics. These are also measures, as they too lack any sense of their dimensionality. The only association that one metric may have with another metric is an association via a power factor, an exponent. None of the metrics shares a fixed zero point on their linear scale with another metric. It thus makes no sense to map on orthogonal axes, as it cannot be decided beforehand that scales are independent. Metrics can show temporarily no relationship with another measure while still being related, like the frozen height in the last regime when the balloon totally occupies the volume of its enclosure. Moreover, the interpretation connected to a metric will change when the system switches to a different regime (or state) where the dependencies are different. For the VF-mapping this means that there is no unique or constant interpretation for the scale value of a metric. For instance, when the voice changes register, a jitter metric or a spectral slope metric could respond with a different sensitivity to an increase in SPL. There is a positive side to this power scalability that will show later when the same power scalability is taken as model for the parameters that together control the voice synthesis. In paragraph 5.9 the idea is explored that the subglottal parameter has no constant proportional (fixed exponential) effect but is also a measure like the balloon height, with a variable exponent that varies as a function of the system geometry.

5.5 Scalability with voice synthesis parameters

Let me here briefly recapitulate the VF-mapping paradigm. Due to the exponential sampling scheme, all voice metrics are on log-scales and are stripped of absolute references to the zero point on their original scale. The control coordinate space becomes an *affine space* [44]. During an interactive VRP-recording, the information on the voice quality that is dynamically sampled along each of the scales of the traced metrics, will travel and vary in concert. Bound by a mutual connection (the system equation and its boundary conditions), each metric traverses its connected scale, each with its own exponential factor. A VRP recording is actually a test for mutual information content, a test for mutual proportional dependencies, a test for shared exponents. This concept, of *analysed* metrics being bound by shared exponents, is thought to reflect a proportional binding that exists between the group of physical and physiological parameters that *together control* the voice.

5.6 Microscale and macroscale in voice synthesis

Many physics-based models of the voice are based on the concept of numerically integrating partial differential equations (PDEs). These microscale equations describe how forces (or force factors using Hamiltonians) are equated within the context of a single small spatial element for which the time behaviour is sampled at a high clock rate [63]. The higher the sampling rate in time, the denser the spatial grid that is used to build a large macroscale structure, the more a microscale multiplication factor will be reiterated. However, with each iteration over time or space, the magnitude scaling effect of such a multiplication factor will grow exponentially. To maintain a constant magnitude scaling effect on the macroscale, all multiplication factors in the model need to be scaled down exponentially. Thus, when integrating from microscale to macroscale, a comparable exponential scaling principle applies as seen in statistical mechanics, only now that microscale and macroscale parameters remain directly related via an exponent, there is no statistical uncertainty to deal with. It is perhaps slightly confusing that the multiplication factors (parameters) from a reiterated equation at the microscale are the direct macroscale controls. For each multiplication factor in the microscale equation, there is the same exponential correction. Thus, it appears that these global macroscale parameters (that are the direct microscale multiplication factors), are together encapsulated within the same exponent. In this situation too, the microscale model is controlled at macroscale via a shared exponential dependency.

The same principle is seen if only the vocal tract is modelled as a filter. With a Laplace transform, a transfer function can be derived by integrating, over the length of the vocal tract, the magnitude and phase change that occurs as a function of the tract geometry. The magnitude (and phase) scaling effects that are associated with the poles and zeroes that characterize the transfer function are again bound in a product relationship. Filtering (a convolution sum, or integral over a micro-timescale) can again be seen as a combined action of multiplicative coupled frequency contributions of resonant and anti-resonant band filters that represent macro-scale parameterized sections. Each parameterized filter section has a proportional (not an additive) effect on the overall power transfer.

5.7 No observable microscale only a real macroscale

Imagine that there is no adequate physical model of the voice available, but only the real thing! With the voice there are only global, macroscale controls, like the subglottal pressure, the muscles that regulate the longitudinal tension and the adduction, the mouth opening, supraglottal area, etc. Following the earlier line of reasoning, the settings for these macroscale physical and physiological parameters, that *together control* the voice, are bound by shared exponents. If these control parameters would not share this type of binding, they are not in control of the same mechanism. The idea that there are two active systems, aerodynamically uncoupled, for which the energy can be independently controlled is just not realistic. How does this translate to voice production in practice?

5.8 Ensemble control

A single physical or physiological control parameter could change by a certain amount and the acoustic system will react proportionally in relation to the scaling not only of this parameter but also of the other control parameters. The voice production mechanism is essentially a *dynamic system*; an oscillating mechanism that is only indirectly controlled by its ‘parameters’. To control the voice, several parameters always act together, each with their own deterministic but sometimes unpredictable, effect. The effect of a single control parameter can be major or minor, according to the state the system is in, but the control is always a team effort, like a ball that is kept in the air by a group of people. It remains one system. An essential insight is that, during phonation, none of the control parameters can exert complete and exclusive control. For instance, putting in extra effort, by increasing the subglottal pressure or the longitudinal tension, will produce a desired effect only if it is co-articulated by the other control parameters. When the system (vocal folds plus air column) vibrates, the internal feedbacks that self-stabilize the oscillating mechanism remain in power; the system will be completely defined by the proportional response to its controls, *none of the parameters is an absolute control*. It is a system that ‘floats’ in a dynamic control space, with no absolute scale zero that might instantly nullify the power. It is an affine control space. This conceptual control space is, in an obscurely transformed way, reflected in the distribution shapes of the analysed metrics that are mapped over the VF. Both parameterized models, the voice control model and the analysis model where voice quality metrics are mapped over the VF, have the property that their parameters are proportionally bound.

Here the term ‘proportional’ is used synonymously with a multiplicative or reciprocal effect. More generally such a dependency can be interpreted as a constant exponential link, a constant (or linearly constant) ratio observed when any two log scales are compared.

5.9 Subglottal pressure as a measure

Let us start with the idea that lung pressure is essentially a volumetric control, as it is a force that can produce work in three different dimensions at the same time. When air is contained in a volume, the air pressure will be in a reciprocal relationship to this volume. As in the balloon example, in voice production mechanism too, pressure and volume remain bound by this reciprocal dependency, where this macroscale relationship remains in action even if we downscale to the millimeter. This tradeoff between pressure and volume constitutes an equating action in which three spatial dimensions are always involved.

Subglottal pressure (P_{sub}), acting on the vocal folds, is able to exert a force in several dimensions. It is not a force that acts exclusively upstream, or sideways. Forces exerted in all three dimensions may increase the energy of the vibrating vocal folds, and with that the total energy contained in the system that includes the air column. The force factors related to P_{sub} will depend on the size and orientation of the vibrating surface and its ability to move in different directions.

As soon as we model (in a physical model) P_{sub} as a macroscale force acting in only one dimension [65][66][67][68][69], then this force component should be considered a *measure* that reflects P_{sub} with a variable exponent. Just as with the balloon example, this exponent depends on the collapsing (or sharing) of force components in the other dimensions. P_{sub} is never a one-dimensional driving force that is exerted on the vibrating vocal folds. In the same manner, the acoustic pressure is caught in a volumetric dependency within the same system. The two pressures, P_{sub} and the acoustical pressure, cannot be expected to simply equate one-to-one. The two decades or so (the exponential factor) over which P_{sub} can vary are thus not indicative for the exponential factor with which the acoustical pressure is able to vary. P_{sub} is essentially a volumetric control that, in current models, is assigned a reduced dimensionality. Similarly, the dimensionality of the acoustical pressure finds is a variable, too. The power contribution factor in dB P_{sub} per dB SPL is the variable to exploit and optimize in voice production, where there are three dimensions (and powers) to equate and benefit from. Note, that here we do not use the concept of a glottis or consider the air flow as an argument in the reasoning; the process that connects the two pressures is not considered in detail.

According to the above line of thinking, P_{sub} is a measure for the acoustical power produced, but never in an absolute sense. P_{sub} only will have a calibrated linear effect when the force distributes in Newton on the microscale, but the exponent for P_{sub} as an integrated, multidimensional force on the macroscale will depend on the physiological properties of the complete system.

The subglottal pressure and volume are bound by an equilibrium of forces with the vocal fold tissue in which also the air column with the supraglottal (tract) pressure and volume are involved. A delicate issue in the above reasoning is, that in this overall system equation, the *acoustic pressure* is typically a very minor force factor. The question about the equilibrium of forces is different from the question how these factors relate to the total energy contained in the system. If it is about energy, forces give a differential (microscale) answer on the amount *change* in energy, but not a (macroscale) answer on the number of Joules contained in the system.

However, the *acoustic pressure*, like the temperature variable in a thermodynamic system, has a double role; it is a factor in the system equation, but its power level reflects on the total energy in the overall system too. How much percent of this energy is actually reflected in the acoustical power that is picked up by the microphone, is not a question we can easily answer by a deduction based on forces. However, the localizing principle can be seen as an empirical confirmation that this percentage must be rather constant. If this transfer factor was that easily varied by for instance a vocal tract adjustment, all metric patterns would rapidly dislocate vertically over the VF, but the effects of vowel changes are not at all that dramatic (see paper **E**). The other scenario is that vocal tract adjustments do change this power transfer factor, but with it, the total energy contained in the vibrating system varies with the same factor. However, the fact that EGG based metric distribution shapes, that more directly reflect on the amplitude of the vocal fold vibration, also obey the localizing principle [17][18][19][70], seems to contradict this last scenario. Then again, the idea that indeed the total energy of a system that includes the masses of the vibrating vocal folds is reflected in the acoustical power that is picked up at a fixed microphone distance, sounds as a precarious concept when the effect of mouth opening and radiation is considered. There are no definite answers at this moment, but the localizing principle does pose a very challenging question.

So, both P_{sub} and acoustic pressure are observable signals (or measures). The subglottal pressure in Pascal does not control the total energy contained in the system in an absolute sense, but the power of the acoustic pressure (SPL) does seems to shadow this energy level rather genuinely. P_{sub} may be the bigger brother that feeds the process, but his power is not absolute.

6 CASE STUDY

The purpose of this example is to demonstrate how individual voice quality change can be monitored in detail. Each voice tells a different story, and voice problems are typically ‘local’, that is, contextual. The scanning of the full range that is a premise of the VF mapping paradigm will assure that ‘local’ problems (and possible solutions) are exposed.

6.1 Case

The female classically trained singing student for which the VF recordings are presented here, experiences a ‘problem’ when she sings in the high part of her voice range. The student is very skilful and motivated to become a professional singer. Due to her talents, she has developed clever compensation techniques to work around her problem and to hide it. Thus, the problem is not immediately clear to an inexperienced listener. A consulted ENT specialist did a video-stroboscopic examination, and documented to have observed a thickened / hardened tissue structure that is situated slightly below the contact area of the vocal folds. In modal voice, this tissue structure does not always show or interfere with the vibration. The goal at this point, is to avoid an invasive treatment and to see if the problem can be solved by voice therapy, an adaptation in life style, specific singing training exercises, an improved vocal hygiene, and other forms of self-treatment. This attempt to improve the voice quality on all fronts will be called the ‘training-period’.

6.1.1 Aims

The aim of the VRP recordings was manifold: (1) to sketch in an objective way the acoustic contours of the problem, (2) to formulate distinct training goals on base of the objective findings, (3) to document the status pre- and post-training, and (4) to demonstrate the applicability of the system to colleagues in the field.

6.1.2 Context

Recordings were made in 2014, by the author PP in cooperation with the singing teacher, who coaches and instructs specific voice exercises. Recording room ca 50 m^3, slightly reverberant. Background noise level ca. 35 dB (A) SPL. Recording system: Voice Profiler 5.5 spectral.

6.1.3 Results

Results were obtained during two recording sessions (pre/post), with a period of 4 months between sessions. Each recording session took about an hour; it is not a routine test. In both sessions, separate VRP recordings were made for the two voice registers; modal/chest voice (M1) was recorded first and falsetto/head voice (M2) second. Additionally, a SRP was recorded by instructing a simple counting task.

6.2 Session I (pre-training), SRP and VRP density distributions for M1 & M2.

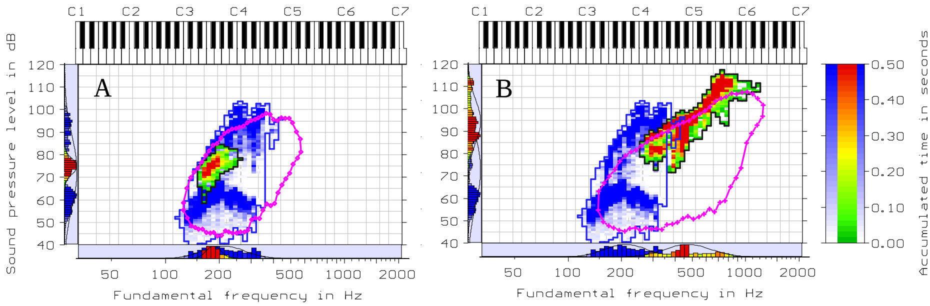

*Figure 16. Pre-training phonation density distributions, with colour scales showing the accumulated time in seconds. In the background (white-blue scale) is the VRP of modal/chest voice (M1) in both panels A and B. Panel A shows in front (red-yellow-green scale) the SRP that is obtained with the counting task. Panel B shows in front the VRP of falsetto/head voice (M2). For reference to a comparable 'normal' population, the pink curves show the FD-based average VRP contours for M1 and M2 respectively, of the group of trained female voices (TF) from Paper **E**. The FD-averaging method is described in paper **C**.*

6.2.1 The threshold of phonation in M1

The thick blue band at the bottom in Figure 16, Panel A has the shape of an inverted V. With this student singing in M1, the phonation threshold, (the L_{min} curve or bottom contour), first goes up and next down with increasing f_o. This upward arch is characteristic to this voice, and reappears in the later post-recording. Like the VRP contour shape, the metric distribution shapes are generally reproducible and personal. The L_{min} curve arching is opposite to the trend seen with the FD-average, but there is no special physiologic explanation why this is happening with this voice. Voices may show different contour curves without a clear reason or without being deviant. It is tempting to attach interpretations, to make more from the VRP contour. Observations on specific aspects of the VRP contour generally make sense during a recording, but are seldom generalizable. When voice quality metrics are brought into the picture, it often turns out that causalities are non-trivial, although they might appear to be so when only a phonation density metric is displayed. To illustrate this, all the density based evidence will be considered here first, while later in paragraph 6.5, the message that two voice quality metrics convey will be added to the picture.

6.2.2 Tonal range and SPL range in M1

Compared to the FD-average that is marked by the pink curve in Figure 16 Panel A, the tonal range in M1 falls short towards high f_o . The FD-average is not intended as a norm, but it does give a hint. During this first recording session, the singer and the teacher decided to stop the search for the maximum SPL at a high f_0. There was a feeling that this would lead to vocal damage. As explained in paper **D**, classically trained female singers focus on the quality of M2, and avoid the use of pushed-up chest voice. Singers may lose this skill. The same paper shows an example of the large range changes that may happen to an individual voice as a result of training and adapting to a classical singing style. So, the impression after the first session was, that this singing student could expand on her M1 area.

6.2.3 Tonal range and SPL range M2

It becomes very evident that there is a problem with this voice when the SPL range and f_o in M2 are assessed (see Figure 16 Panel B). The singing student is capable to sing very loudly over a range of at least two octaves. She even reaches L_{max} values that go above the extremes that were observed with the TF group from paper **E** (the pink curve only shows the FD-contour average and not the union VRP area). The singing student is unable to sing softly in M2. She is still able to perform on stage, as long as she sings very loudly. Typically, female voices become breathy when, at high f_o, the sound level goes to minimum. With her voice, there is a breathy quality even at high SPL and when the level is lowered, the vibration just stops.

6.3 Training goals

While taking into account the result from this first recording session, the teacher formulated several training goals. The main goal, was to improve the quality and control in midrange. Moreover, vocal loading should be reduced. Singing exercises should focus on equalisation of the quality in M2. Around A4 (f_o=440 Hz) the range over which the SPL could be varied finds a maximum (see Figure 16 Panel B). The singing student had already acknowledged this setting as her best tone. Training exercises that aim at expanding the dynamic range in M2, should logically work from this centre. It is singing, not rocket science.

6.4 Session II (post-training), SRP and VRP density distributions for M1 & M2.

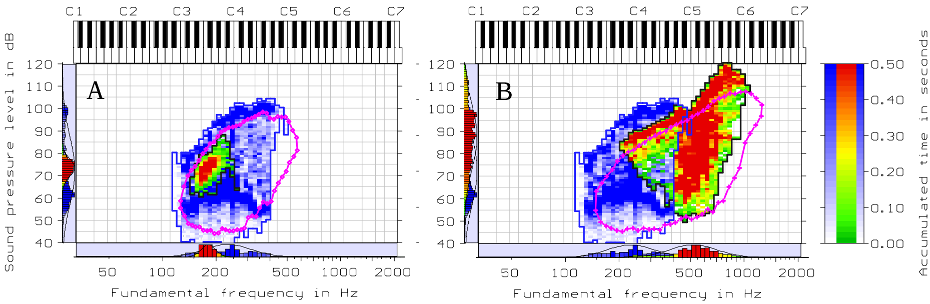

Figure 17. Post-training phonation density distribution, for comparison with Figure 16. Panel A shows in front (red-yellow-green scale) the SRP that is obtained with the counting task, while panel B shows in front the VRP of falsetto/head voice (M2). In the backgrounds (white-blue scale) is the VRP of modal/chest voice (M1). The pink curves again show the FD-based average VRP contours for M1 and M2 from the reference population.

6.4.1 Tonal range and SPL range in M1, post-training.

Figure 17, Panel A shows that post training, the L_{max} contour in M1 follows the curve of the average L_{max} FD-contour at a distance of about 7 dB. The spacing remains rather constant over the full f_o range from A2 to C5. Note that the contour curves align in the north-west corner and thus match the general finding discussed in Paper **C**. Above C4, the area between L_{min} and L_{max} curves is the now filled-in with a blue colour. This blue filling was not seen in the pre-training VRP (c.f. Figure 16 Panel A). This filling means that, during the post-recording, the singing student was able to do *messa di voce* exercises from L_{min} to L_{max} up to the highest tones in her M1 range. This inner area coverage is a documentation of the improved control in ‘midrange’ that was part of the training goal. Note that the tonal area that referred to as ‘midrange’ in VRP terms would also be called the ‘register overlap zone’, the tonal range that is shared by M1 and M2. Note also that mixing register qualities does not mean that a new intermediate register is created. Both M1 and M2 remain connected to distinct vibratory states and VRP areas.

6.5 Voice quality metric distributions pre/post training

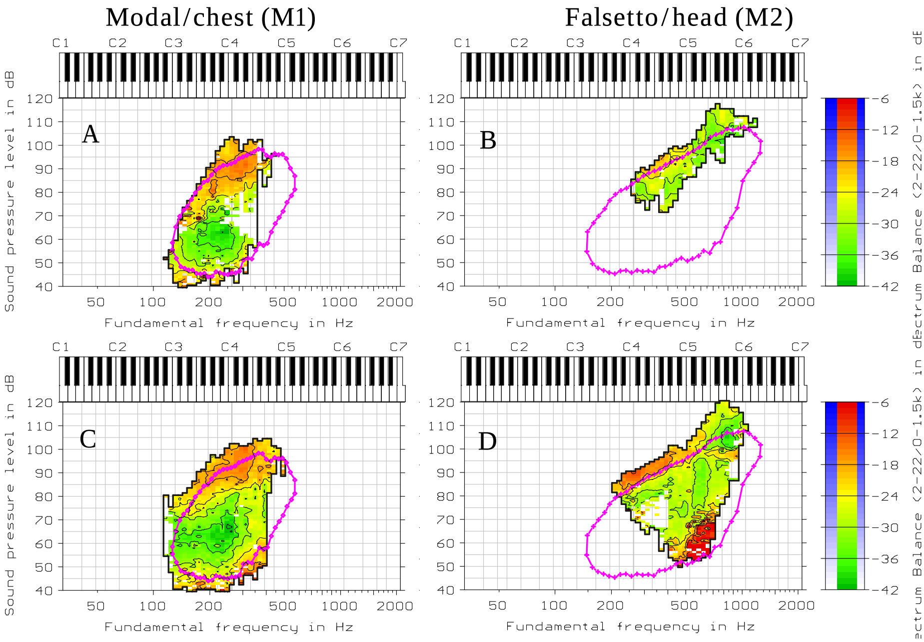

Figure 18. Spectrum balance (SB) distributions. Top row (panels A & B) pre-training, bottom row (panels C & D) post training. Left-hand column (panels A & C) modal/chest voice (M1), right-hand column (panels B & D) falsetto/head voice (M2). Pink curves as in Figure 16.

6.5.1 SB-distributions pre/post

Note that there is no norm presented with the SB metric distributions in Figure 18. The union VRP for the group of trained female voices, presented in paper **E**, could be taken as a reference or template for what to expect. The graph is not reproduced in this Chapter; Figure 8 in paper **E** shows the union VRP maps for the SB metric. It would be misleading to call this average SB distribution a norm (and showing it again), because the SB is not a definitive measure of the quality of a singing voice. The SB metric can be seen as relative measure of the level of the singer’s formant (for reasons explained in paper **E**). There are strong voices that are not characterized by high SB values. The SB is neither a universal, nor an absolute measure of the quality of a singing voice. Such an assumption is too large a simplification, although it looks as if the singer’s formant and the VRP naturally belong to together [71][20][72]. Moreover, as explained in detail in paper **E**, there

are systematic differences between voices in the way SB varies as a function of SPL and f_o. Chapter 5 was concerned with presenting an explanation for why voices do not necessarily comply to the same SB distribution shape. Despite the differences *between* voices, metric distribution shapes *within* a voice are generally reproducible. This in-voice reproducibility is the central premise on which this thesis builds (see Chapter 1). The similarity in Pre/Post SB-distribution shapes for M1 (compare Figure 18 panels A & C) clearly demonstrate the reality of this premise. For an impression of how different the SB distribution shapes are between voices, the reader may explore Charts S7 in the supplementary files of paper **E**, where the individual SB maps for 35 voices are charted for both M1 and M2.
Metric distribution shapes within a subject are expected to change as result of training a different singing technique. In this example, the singing student did training exercises that aimed at equalising (and enforcing) the quality of the voice in midrange while using M2 only. Paper **D** gives a more extended characterization of this singing technique and how it relates to the idea of mixing. Figure 17B shows an extension downward (in f_o) and upward (in SPL) of the VF area in M2 post-training, as compared to the area established pre-training (see Figure 16B). Figure 18D shows the change in the SB-metric distribution for M2 post-training, where Figure 18B shows the pre-training distribution. First note that post-training, the L_{max} contour of the singing student extends at least 7 dB above the average L_{max} FD-contour of the TF group (pink line). Between these two L_{max} curves, over an octave that ranges from A3 to A4, an orange-coloured band of more or less constant SB is observed in the post-training distribution. This band of constant SB is interpreted as a sign of vocal stability and steadfast control of the quality of the voice that results from the equalising training.

So far, the outcome seems positive for the singing student, as the effect of the whole package of actions that we call training clearly payed off, and was confirmed by the objective findings. However, this impression will change when the jitter metric distribution patterns are taken into account.

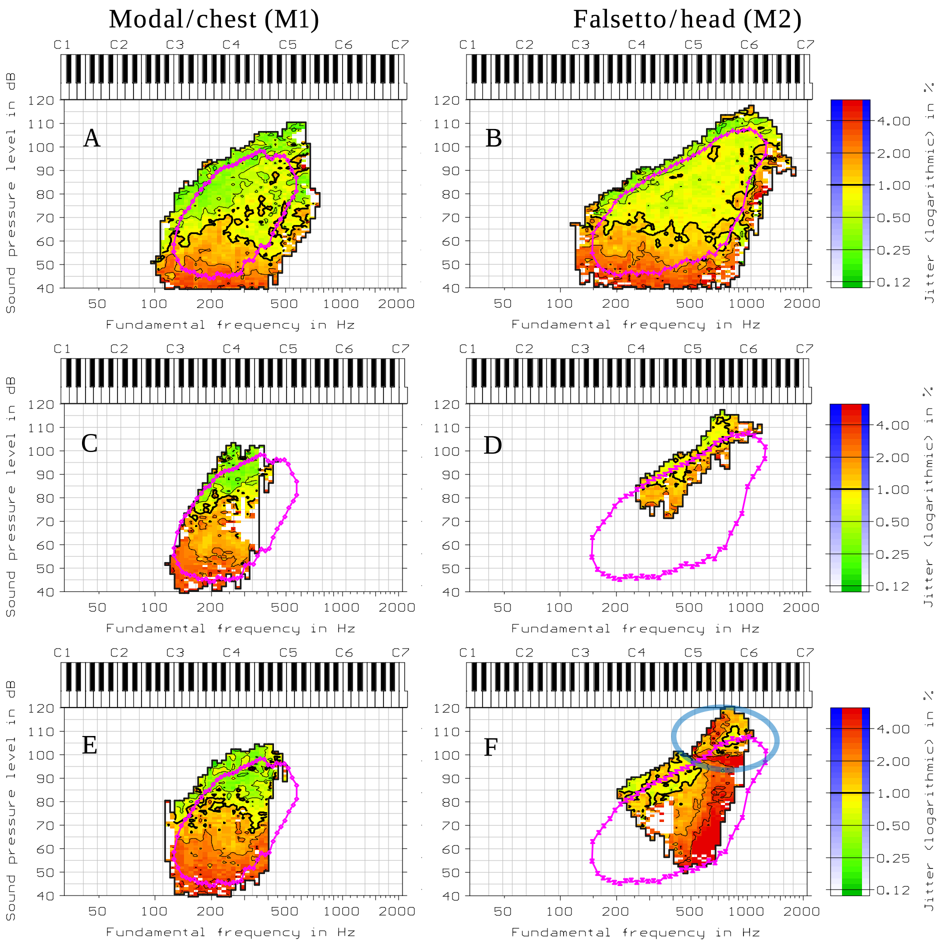

*Figure 19. Distribution shapes for the jitter metric. A and B: Union VRPs for the group of trained female voices (TF) from Paper **E**. C and D: pre-training, E and F: post training. Left-hand column (panels A, C and E) modal/chest voice (M1), right-hand column (panels B, D and F) falsetto/head voice (M2). Pink curves as in Figure 16. The blue oval in panel F marks a high jitter zone that is commented on in the text.*

6.5.2 Jitter distribution Pre/Post

On visual inspection, the jitter distribution shapes pre/post from Figure 19 C/E (M1) show comparable values for the same f_o and SPL.

Note that there are ways that are more objective to judge a similarity in VF distribution shape. However, the standard solutions not automatically apply. As mentioned before, a standard deviation has a different statistical interpretation on a logarithmic scale (see Chapter 4). A more elaborate treatment of how to compare VF distribution shapes objectively is not included in this thesis. Note that the phonation density distribution shape (Figure 16) does not shine through in the jitter distribution shapes seen in Figure 19. Here too, a visual pattern comparison is for now the practical solution.

Panel A in Figure 19 shows the average jitter distribution shape for the TF group from paper **E** (this figure does not occur in paper **E**). Note the decrease in jitter with increasing SPL, that was already discussed in papers **A** & **B**. With our singing student, for a given SPL, higher jitter values are found than for the average trained female voices. We could thus conclude that the jitter was raised for the voice in this case study, as there was obviously a more irregular vocal fold vibration both pre- and post training. However, this is a dangerous conjecture.

> The jitter metric for which the distributions are displayed here, measures a form of irregularity that according to the detection scheme can officially be named 'jitter'. In the Voice Profiler system however, there are several parallel metrics implemented that all sense jitter with a slightly different detection scheme. These metrics are not shown to the customers, but they assist in tracking the f_0. The actual jitter metric that is displayed here does not sense selectively the predominantly low frequency aperiodicity that is associated with our perceptual impression of roughness of the voice. With this more wide-ranging aperiodicity metric, the VF distribution shapes are more reproducible within individuals. Note, that the locating principle is liberal (see paragraph 1.4), it will guarantee reproducibility for any metric that reflects on the voice. Reproducibility and system independency are important attributes of a metric. Tests with a synthetic voice source (see paragraph 2.4.1), demonstrated that an increased period-to-period irregularity does show as an increased 'aperiodicity value' with the displayed metric. So, the implemented metric does sense the targeted aperiodicity aspect and follows the classical detection scheme (see paper **A**). However, an irregular vocal fold vibration may perturb the acoustic waveform in several ways. From this overall perturbing effect on the waveform, different remnants can be traced using different analysis models. Moreover, many jitter detection schemes will sense an additional noise component too, that are typical to a breathy voice

quality. Different forms of aperiodicity are not simply separable. Schoentgen [73][74][75] treats these issues with aperiodicity in depth, and with synthesis examples [76] demonstrates his mastery in this matter. My customers will not all read his work and follow his level of abstraction. Users of the system will not automatically recognize that there are many expressions of the same phenomenon with a multiplicity of metrics and analysis models that all gather under the ‘jitter’ flag. There is a huge literature on this subject that was consulted at an early stage in the development of the different recording systems, and that is not referenced in this framework, by many brilliant researchers to whom I remain indebted.

In the main text above, jitter is treated as a single phenomenon, linked to a single abstraction, which reality proves to be a gross simplification. For the Voice Profiler implementation, the wide ‘jitter metric’, selected as a flagship, senses a combination of aperiodicity aspects. With a healthy voice, this metric gives a slightly exaggerated impression of the irregularity. Note that there is a lot of irregularity or uncertainty in the sound signal that our ears do not take into account [77]. This exaggerated impression is a problem mainly in M1. The bonus of this lack of specificity is that the ‘jitter metric’ will be very sensitive to any small form of aperiodicity at high f_o in M2. This is a desirable property. The resulting clear link to the perceived breathiness and roughness at high f_o helps to convince the customers of the relevance of the additional quality measurement. Many customers were sceptical of doing acoustical voice quality measurements. They had experienced a weak correspondence between the perceived quality and the measured jitter, before, when using other acoustic voice analysis software.

So, the three jitter distributions for M1 (see Figure 19 A, C & E), do not reflect only on the roughness of the voice, but report on an overall irregularity, that is not necessarily of a pathological origin. As with the earlier SB metric, jitter distribution shapes are characteristic to individuals. A jitter around 1% is normal to a voice (note that the jitter scale is not universal but depends on the metric). A jitter above 1% over half the VF area, as observed with singing student in M1, does not directly reflect on a lesion that affects the overall periodicity of the vibration mechanism.

In this case study, the effect of the stroboscopically observed impediment is much more local. The effect shows very explicitly as a red coloured zone in M2 (Figure 19 panel F). An extreme jitter above 4% was recorded post-training in M2, at sound levels above 90 dB SPL (marked by a blue oval). The red stain unmistakably confirms the perturbed voice quality that was directly heard during the interactive recording. The marked horizontal red stroke, that from pitch B4 to B5 gradually runs down from 100 dB SPL to 95 dB SPL, is a phenomenon that is seldom observed. Voice dysfunction generally goes hand in hand with a reduction in maximum SPL. Here, that is not the case. During the post-training session, the singing student is singing extremely loudly and her voice remains very powerful over the full range, despite the problem. In an odd way, all the training paid off, as it improved the control and overall quality of her voice. It did not solve the problem, but made it more distinct.

6.6 Qualitative or quantitative, contour or quality metrics?

The above interpretation is largely qualitative, while building on quantifiable results. Note that there is no number specified to rate the increase in pre/post VRP area for M1 or M2. Such numbers would give a general impression but miss the essence. Although more area is gained with M1 and M2, and the control post-training improved markedly, the voice quality actually worsened in a small, but critical zone. All over M2, the voice became breathier and the periodicity of the vibration became more perturbed. As these findings concern particulars in specific quality distribution shapes, there is no point in searching for a further generalisation. Note that it is the voice quality metrics and not the VRP contour that present the most relevant message. However, the import of the quality metrics became clear only thanks to the goal to explore the full VF area.

6.7 Conclusions

What the example demonstrates, is that the numerical values for a voice metric make sense within a context. A difference on a jitter or SB scale becomes meaningful thanks to this contextualisation. The sense of detail on a scale depends on the context too. Voice changes as a result of voice training or therapy, are linked to a localized context. Normative values could be specified for a voice metric (for instance, jitter below 1%, an SB above -10 dB). Such norms, however, lose their role of reference in a personal context. It is doubtful if the idea of specifying absolute norms, which deny this contextual factor, will ever be realistic. The framing of a context, as with the VF mapping paradigm, may become even more important than the actual numerical values of the metrics. This does not imply that the actual numerical values do not matter anymore.

Metric values remain part of a spatial VF distribution shape, and this distribution is generally different between voices. Perhaps the only generalization that is valid for all voices and all voice metrics is that they all comply to the localization principle. In its turn, due to this localization principle, it becomes evident how difficult it is to make further generalizations.

Voice problems are typically contextual, and they can be exposed by navigating to a specific VF location that refers to this context. In the above example, a high jitter value clearly marked the problem. It was not the wide, red coloured jitter distribution shape that was observed in modal voice that depicted the actual problem. A small red coloured high jitter band in M2 marked a persistent highly perturbed voice quality that appeared only within a narrow range in SPL and f_o . This feature identified the problem for the singer. Numbers became *substantial evidence* due to their embedding within the context.

The ‘keyhole approach’ as ST named it in paper **F**, indicates the problem with taking a narrow selection from the total VF distribution shape to be a representative value. With that approach, there is of course always some possibility of sampling a meaningful contrast. The singing student from the present case study could have been asked to sustain the vowel /a/ at a comfortable pitch and level using modal voice, where next an average jitter could be assessed. A pre-post training comparison of a single ‘characteristic’ jitter value might have led to a result that looks meaningful. There could even be a causal relationship to the problem, but it would not necessarily be the sought causality.

The keyhole approach is not wrong, but it is a different assessment paradigm, a different test model. With that model, it is questionable if the assessed numbers will ever become interpretable in a useful way, since the inherent (de)contextualisation plays such a crucial role. The discussion of the VF mapping paradigm in this thesis gives an idea of what the confounding, unknown variables might be, with the keyhole approach.

We like simple, controllable protocols that take only ten minutes. We like clever abstractions and numbers that speak for themselves. We do not want to be forced to answer with each test the question: ‘what am I looking at?’. Why scan the whole voice range?

Yet spending an hour, instead of ten minutes, and assessing a wealth of information linked to a VF context can be well worth the trouble, especially if it means getting objective information where on which one can build therapy and training. The question of which approach will eventually pay off more, and will help in solving a specific problem of a voice patient or singer, does have an answer.

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